

Excel, AI & Power BI

for Data Analytics and Business Productivity

Election Analytics Simulation

A fictional case study: National Election Analytics Simulation 2026

2-Day 1-Night Practical Training Programme

28 – 29 September 2026

TRAINER

Fakhrul Syahmi

Data Consultant & HRD Corp Certified Trainer
fakhrulsyahmi@mufasyahconsultant.com

All data in this handbook is fictional and created solely for training purposes.

PARTICIPANT HANDBOOK

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WELCOME

Before You Begin

IMPORTANT | Fictional Data Disclaimer

All election data, political parties, candidates, constituencies, results, voter profiles and scenarios used in this training material are completely fictional and have been created solely for educational and data analytics training purposes. They do not represent any actual election, political party, candidate or constituency.

Every number in this handbook was produced by a random data generator. Any resemblance to real places, organisations or outcomes is coincidental. The focus of this programme is **data analytics technique**, not political commentary.

About This Handbook

This handbook is your companion for the two-day programme. It follows one complete analytics journey: you start with a messy election dataset, clean and explore it in Microsoft Excel, use Generative AI to speed up analytical work, and finish by building an interactive dashboard in Microsoft Power BI.

You will work with **one consistent fictional election universe** from the first module to the last. The same parties, constituencies, candidate codes and election years appear in every exercise, so the skills you build in Excel carry directly into Power BI.

How to use this handbook

Element	What it means
Module	A self-contained topic with explanation, worked examples and exercises.
HANDS-ON EXERCISE	A task you complete on your laptop using the training files.
CHECK YOUR ANSWER	The expected result, taken from the actual training dataset. Use it to confirm your work.
TRAINER TIP / BEST PRACTICE	Practical advice from real-world analytics projects.
COMMON MISTAKE	Errors participants often make, and how to avoid them.
AI PROMPT	A prompt you can copy into ChatGPT, Claude, Copilot or Gemini.

Training files

You will receive the following files. Save them in one folder, for example `C:\ElectionTraining\`, before the session starts.

File	Contents	Used in
Raw Election Data (Messy).xlsx	Two sheets with deliberate data quality problems: RawElectionResults and RawPollingStations2026	Modules 2, 14; Capstone
Election Results.xlsx	Clean candidate-level results, 269 records across 2022 and 2026	Modules 3–7, 13–18
Constituency Master.xlsx	40 constituencies with voter profile and turnout	Modules 4, 6, 13–18
Candidate Master.xlsx	Candidate attributes by election year	Modules 4, 13, 15
Party Master.xlsx	5 fictional parties and codes	Modules 4, 13, 15
Polling Station Results.xlsx	719 polling-station records across both elections	Modules 5, 13–18

Learning Outcomes

By the end of this programme, you will be able to:

1. Explain the structure of election data and identify common data quality problems.
2. Clean and validate a messy dataset using Excel functions and tools.
3. Calculate turnout, vote share, majority, vote change and rejected-vote rate correctly.
4. Summarise results with PivotTables, PivotCharts and slicers, and build an Excel dashboard.
5. Write structured prompts that make Generative AI useful for formulas, checks and summaries.
6. Verify AI output against the data before using it.
7. Import, transform and model related tables in Power BI using a star schema.
8. Write DAX measures that respond correctly to filter context.
9. Build a four-page interactive election analytics dashboard.
10. Present descriptive, data-supported findings and recommend further analysis.

2 DAYS · 1 NIGHT

Programme Schedule

The programme runs over two days with one evening session. Timing is a guide; the trainer may adjust it to suit participants' pace.

Day 1 — Monday, 28 September 2026: Microsoft Excel

Time	Session	Focus
8:30 – 9:00	Registration & Briefing	Setup check, training files, disclaimer
9:00 – 10:30	Modules 1–2	Understanding election data; cleaning data in Excel
10:30 – 10:45	Morning Break	
10:45 – 12:30	Modules 3–4	Formulas for analytics; lookup functions
12:30 – 2:00	Lunch & Prayer	
2:00 – 3:30	Modules 5–6	Election KPI calculations; PivotTable analysis
3:30 – 3:45	Afternoon Break	
3:45 – 5:15	Module 7 + Challenge 1	Excel election dashboard
5:15 – 5:30	Day 1 Recap	Questions and reflection

Night Session — Monday, 28 September 2026: Generative AI

Time	Session	Focus
8:30 pm – 9:15 pm	Modules 8–9	Understanding Generative AI; prompt engineering
9:15 pm – 10:00 pm	Modules 10–11	AI for Excel; AI-assisted analysis
10:00 pm – 10:30 pm	Challenge 2	Prompt-writing challenge

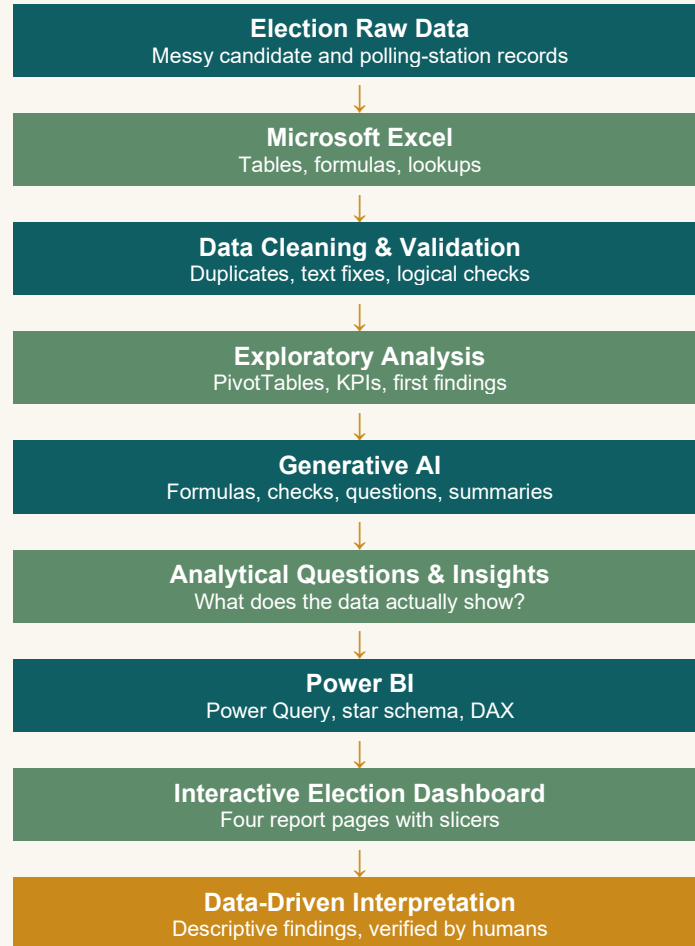
Day 2 — Tuesday, 29 September 2026: Microsoft Power BI

Time	Session	Focus
8:30 – 10:30	Modules 12–15	Power BI overview; connecting data; Power Query; data model
10:30 – 10:45	Morning Break	
10:45 – 12:30	Modules 16–17	DAX measures and advanced election metrics
12:30 – 2:00	Lunch & Prayer	
2:00 – 3:30	Module 18 + Part 4	Building the dashboard; AI + Power BI
3:30 – 3:45	Afternoon Break	
3:45 – 5:00	Capstone & Challenge 3	Raw data to interactive dashboard; group presentations
5:00 – 5:30	Closing	Key takeaways, evaluation, certificates

THE ANALYTICS JOURNEY

Training Roadmap

Every exercise in this programme fits into one workflow. Raw data enters on the left, and a data-driven interpretation comes out at the end. Each tool has a clear job.





Which tool does what?

Tool	Main role in this programme	Typical output
Microsoft Excel	Prepare and explore data	Clean table, KPI calculations, PivotTables, quick dashboard
Generative AI	Accelerate analytical tasks and problem-solving	Formulas, DAX drafts, data quality checklists, question lists, summaries
Microsoft Power BI	Model, visualise and communicate data	Star schema, DAX measures, interactive multi-page report
You (the analyst)	Validate, interpret and decide	Verified findings and recommendations

CENTRAL CASE STUDY

The Case Study: National Election Analytics Simulation 2026

You have joined the data team of a fictional national election commission. Two general elections have taken place in the fictional country: **Election 2022** and **Election 2026**. Your job is to turn the raw results into reliable numbers and an interactive dashboard that answers descriptive questions such as *Where was turnout highest?*, *How did vote share change?* and *Which contests were closest?*

The fictional universe at a glance

Item	Value in the dataset
Election years	2022 and 2026
States	8 fictional states in 4 regions
Constituencies	40 (5 per state), classified Urban, Semi-Urban or Rural
Parties	Party Alpha, Party Beta, Party Gamma, Party Delta, Independent
Unique candidates	161 (some contest both elections)
Candidate-result records	269 (136 in 2022, 133 in 2026)
Polling-station records	719 (355 in 2022, 364 in 2026)
Registered voters (2026)	3,171,746

States and regions

Region	State	State code	Constituency codes
Northern	Northaven	NHV	NHV-01 to NHV-05
Northern	Highmoor	HMR	HMR-01 to HMR-05
Eastern	Eastmere	EMR	EMR-01 to EMR-05
Eastern	Coastgate	CGT	CGT-01 to CGT-05
Southern	Southvale	SVL	SVL-01 to SVL-05
Southern	Lakeshire	LKS	LKS-01 to LKS-05
Central	Midfield	MDF	MDF-01 to MDF-05
Central	Central Ridge	CRG	CRG-01 to CRG-05

Fictional parties

PartyCode	PartyName	PartyGroup	ShortName	Candidate ID prefix	Chart colour
PA	Party Alpha	Grouping X	Alpha	A (e.g. A001)	Teal #3A7D7C
PB	Party Beta	Grouping Y	Beta	B (e.g. B001)	Amber #C98A1B
PG	Party Gamma	Grouping X	Gamma	G (e.g. G001)	Slate #6C7A89
PD	Party Delta	Grouping Y	Delta	D (e.g. D001)	Mauve #8C6A8F
IND	Independent	Non-aligned	Indep.	I (e.g. I001)	Grey #9A9A9A

The chart colours are deliberately neutral. They are not associated with any real political organisation. Use them consistently in Excel and Power BI so that each party keeps the same colour on every chart.

The worked example: Ashford Vale (NHV-01)

Throughout the handbook we return to one constituency so you can see every calculation on the same numbers. **Ashford Vale (NHV-01)** is an Urban constituency in Northaven, Northern region. In Election 2026 it had 118,954 registered voters, 83,589 ballots issued, 82,291 valid votes and 1,298 rejected votes.

CandidateID	Party	VotesReceived	Rank	ResultStatus
A001	Party Alpha	37,950	1	Won
B001	Party Beta	22,417	2	Lost
G001	Party Gamma	15,654	3	Lost
D001	Party Delta	6,270	4	Lost

Check: $37,950 + 22,417 + 15,654 + 6,270 = 82,291$, which equals the valid votes. This is the first data quality rule you will learn: **candidate votes in a constituency must add up to the valid votes.**

DATA OVERVIEW

The Five Training Datasets

The data is stored in five related tables. Each table has one clear purpose and one clear level of detail, called its **grain**. Understanding grain is the key to avoiding double counting later in Power BI.

Table	Grain (one row = ...)	Key column(s)	Rows
Election Results	One candidate in one constituency in one election	ElectionYear + ConstituencyCode + CandidateID	269
Constituency Master	One constituency	ConstituencyCode	40
Candidate Master	One candidate in one election	CandidateID + ElectionYear	269
Party Master	One party	PartyCode	5
Polling Station Results	One polling station in one election	ElectionYear + PollingStationCode	719

Dataset 1 — Election Results (columns)

Column	Type	Description
ElectionYear	Whole number	2022 or 2026
State / Region	Text	Fictional state and region
ConstituencyCode	Text	Format SSS-NN, e.g. NHV-01
ConstituencyName	Text	e.g. Ashford Vale
ConstituencyType	Text	Urban, Semi-Urban, Rural
CandidateID / CandidateName	Text	e.g. A001 / Candidate A001
PartyCode / PartyName	Text	e.g. PA / Party Alpha
VotesReceived	Whole number	Votes for this candidate
ValidVotes	Whole number	Total valid votes in the constituency (repeats on every candidate row)
VoteSharePct	Decimal (%)	VotesReceived ÷ ValidVotes
ResultStatus / Rank	Text / Whole number	Won or Lost; 1 = winner
Majority	Whole number	Winner votes – runner-up votes (winner row only; blank for others)
PreviousElectionVotes	Whole number	Same party's votes in the same constituency in 2022 (2026 rows only)
VoteChange / VoteSwingPct	Whole number / Decimal	Change in votes; change in vote share (percentage points)

COMMON MISTAKE | Summing a repeated column

ValidVotes repeats on every candidate row. Ashford Vale has four candidates in 2026, so summing ValidVotes in Election Results gives $4 \times 82,291 = 329,164$ instead of 82,291. Always take turnout, valid and rejected totals from **Polling Station Results**, which has one row per station.

Dataset 2 — Constituency Master (columns)

ConstituencyCode, ConstituencyName, State, Region, ConstituencyType, RegisteredVoters (2026), MaleVoters, FemaleVoters, Age18_29, Age30_44, Age45_59, Age60Plus, PreviousTurnoutPct (2022), CurrentTurnoutPct (2026), TotalPollingStations (2026). All voter-profile numbers are randomly generated.

Dataset 3 — Candidate Master (columns)

CandidateID, CandidateName, PartyCode, PartyName, AgeGroup, CandidateCategory, IncumbentStatus (New, Incumbent, Re-contesting), ElectionYear, ConstituencyCode. A candidate who contests both elections appears twice, once per year.

Dataset 4 — Party Master (columns)

PartyCode, PartyName, PartyGroup, ShortName. This small table is the single source of truth for party names.

Dataset 5 — Polling Station Results (columns)

ElectionYear, State, ConstituencyCode, PollingStationCode (e.g. NHV-01-S03), PollingStationName, RegisteredVoters, BallotsIssued, ValidVotes, RejectedVotes, PartyAlphaVotes, PartyBetaVotes, PartyGammaVotes, PartyDeltaVotes, IndependentVotes, TurnoutPct. Party columns hold 0 where the party did not contest.

NOTE | Built-in consistency

The clean dataset obeys these rules, which you will test during the course:

- $\text{RegisteredVoters} \geq \text{BallotsIssued} \geq \text{ValidVotes}$
- $\text{BallotsIssued} = \text{ValidVotes} + \text{RejectedVotes}$ (every polling station)
- Sum of party votes at a station = ValidVotes at that station
- Sum of station votes for a party = that party's VotesReceived in Election Results
- Rank 1 = winner; Majority = winner votes – runner-up votes

PART 1

Microsoft Excel for Election Data Analytics

Excel is where most analysis begins. In Part 1 you will turn a messy election file into a clean, trustworthy table, calculate the core election KPIs, summarise results with PivotTables and build your first dashboard.

- Module 1** Understanding Election Data
- Module 2** Cleaning Election Data in Excel
- Module 3** Excel Formulas for Election Analytics
- Module 4** Lookup Functions
- Module 5** Election KPI Calculations
- Module 6** PivotTable for Election Analysis
- Module 7** Excel Election Dashboard

MODULE 1

Understanding Election Data

Before cleaning or analysing anything, you need to understand what the data represents. Election data looks simple, but small misunderstandings about structure lead to large errors in results.

1.1 Structured data

Structured data is data arranged in a consistent table: every row follows the same layout, and every column holds one kind of information. Election results are a classic example. A spreadsheet of candidate results, a list of constituencies and a polling-station log are all structured data.

Unstructured data has no fixed layout, for example a press statement, a scanned tally sheet or a social media post. This programme works only with structured data.

1.2 Rows, columns, records and fields

Term	Meaning	Election example
Row / Record	One complete observation	Candidate A001's result in Ashford Vale, Election 2026
Column / Field	One attribute measured for every record	VotesReceived
Value	The content of one cell	37,950
Header	The field name in the first row	ConstituencyCode
Key	A field that uniquely identifies a record or links tables	CandidateID, ConstituencyCode

In a well-formed analytics table, **one row is one thing** and **one column is one attribute**. Merged cells, subtotal rows and blank spacer rows inside the data break this rule and cause problems in PivotTables and Power BI.

1.3 Data types

Every field has a data type. Excel and Power BI treat each type differently: you can sum numbers, but not text; you can sort dates in time order, but only if they are real dates.

Data type	Examples in our dataset	What you can do with it
Text	ConstituencyName, PartyName, State	Group, filter, count, label
Whole number	VotesReceived, RegisteredVoters, Rank	Sum, average, compare, rank
Decimal / Percentage	VoteSharePct, TurnoutPct	Average (carefully), compare, format as %
Date	ElectionDate (in the raw file)	Sort chronologically, extract year
Code (text that looks like a number)	PollingStationCode NHV-01-S03	Treat as text; never sum

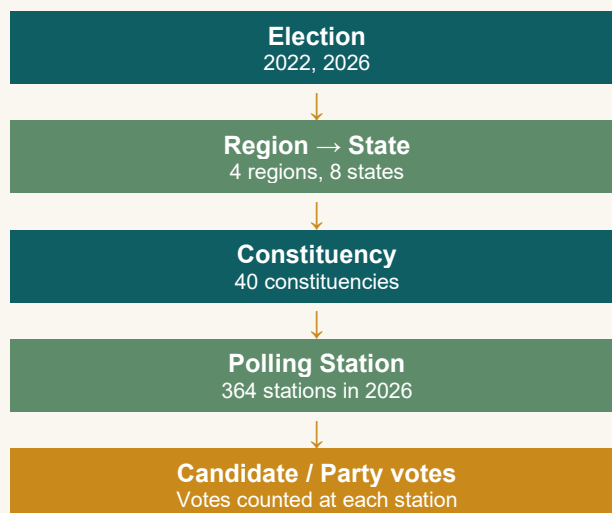
1.4 Categorical and numerical data

Categorical data places records into groups: State, Region, ConstituencyType, PartyName, ResultStatus. You analyse categorical data by counting and grouping. **Numerical data** measures quantities: VotesReceived, RegisteredVoters, BallotsIssued. You analyse numerical data by summing, averaging and comparing.

Some fields are numbers stored as categories. **ElectionYear** is a good example: you filter by it and compare years, but you never add 2022 + 2026. **Rank** is an ordered category: 1 is better than 2, but averaging ranks rarely means anything.

1.5 How election data is structured

Election data naturally has several levels of detail. Understanding the levels tells you where each number should come from.



1.6 Key election fields explained

Field	Plain-English meaning	Ashford Vale 2026
Constituency	An electoral area that elects one representative	Ashford Vale (NHV-01)
Candidate	A person contesting a seat	Candidate A001
Party	The fictional party the candidate represents	Party Alpha

Field	Plain-English meaning	Ashford Vale 2026
Registered Voters	People eligible to vote in the constituency	118,954
Ballots Issued (Votes Cast)	Ballot papers given to voters who turned up	83,589
Valid Votes	Ballots correctly marked and counted	82,291
Rejected Votes	Ballots not counted (spoilt, unmarked, unclear)	1,298
Votes Received	Valid votes for one candidate	37,950 (A001)
Turnout	Share of registered voters who cast a ballot	70.27%
Vote Share	Share of valid votes a candidate received	46.12%
Majority	Winner's votes minus runner-up's votes	15,533

1.7 Garbage In → Garbage Out

Garbage In, Garbage Out (GIGO) means that analysis can never be better than the data it uses. A perfect formula applied to a flawed table produces a flawed answer, and often it looks completely believable.

Here is how three small problems in our raw file would distort the analysis if nobody cleaned them:

Data problem	What happens	Consequence
"PARTY ALPHA", "PartyAlpha" and "Party Alpha" all appear	A PivotTable shows three separate rows for one party	Party Alpha's total is split and understated; ranking may change
A candidate row appears twice	Votes for that candidate are counted twice	Total votes and vote share are overstated; the sum of candidate votes exceeds valid votes
VotesReceived stored as text "12,345"	SUM and SUMIFS ignore the cell	Totals are silently too low; no error message appears
ValidVotes summed across candidate rows	Valid votes multiplied by the number of candidates	Ashford Vale would show 329,164 valid votes instead of 82,291

BEST PRACTICE | Trust, then verify

Before any analysis, ask three questions: **Is every record here once? Is every value the right type? Do the totals reconcile?** In election data, reconciliation is powerful because the numbers must obey arithmetic rules (for example, valid + rejected = ballots issued).

MODULE RECAP

- One row = one record; one column = one field.
- Know the data type and whether a field is categorical or numerical.
- Election data has levels: election, state, constituency, station, candidate.
- Poor data quality produces believable but wrong results (GIGO).

MODULE 2

Cleaning Election Data in Excel

Open **Raw Election Data (Messy).xlsx**. The sheet **RawElectionResults** has 275 rows and contains 184 deliberate problems. In this module you will find and fix them using Excel's built-in tools and text functions.

The raw sheet deliberately leaves out calculated columns (VoteSharePct, Rank, Majority). You will rebuild them in Module 5 once the data is clean.

ElectionYear	ElectionDate	State	Region	ConstituencyCode	ConstituencyName	ConstituencyType	CandidateID	CandidateName	PartyName	VotesReceived	ValidVot
2026	20/06/2026	Midfield	Central	MDL-03	Ivy Crossing	Urban	G030	Candidate G030	Party Gamma	19357	
2026	20/06/2026	Eastmere	Eastern	EMR-1	Kestrel Bay	Urban	A012	Candidate A012	Party Alpha	16848	
2022	18/06/2022	Southvale	Southern	SVL-04	Yarrow Heath	Rural	A025	Candidate A025	Party Alpha	15637	
2022	18/06/2022	Southvale	Southern	SVL-04	Yarrow Heath	Rural	D021	Candidate D021	Party Delta	8210	
2022	18/06/2022	Lakeshire	Southern	LKS-01	Bluewater	Urban	A027	Candidate A027	Party Alpha	19663	
2022	18/06/2022	Midfield	Central	MDL-04	Jasper Hollow	Rural	A037	Candidate A037	Party Alpha		
2022	June 18, 2022	Coastgate	Eastern	CGT-02	Quarry End	Rural		Candidate B018	Party Beta	15,270	
2022	18/06/2022	Coast Gate	Eastern	CGT-04	Saltmarsh	Semi-Urban	A020	Candidate A020	Party Alpha		17358
2026	20/06/2026	Eastmere	Eastern	EMR-02	Larchmont	Semi-Urban	G011	Candidate G011	Party Gamma	10,531	
2022	18/06/2022	Highmoor	Northern	HMR-04	Ironbrook	Rural	D008	Candidate D008	Party Delta		6210
2022	18/06/2022	Northaven	Northern	NHV-05	Elmstead	Urban	A005	Candidate A005	Party Alpha		29783
2026	June 20, 2026	Central Ridge	Central	CRG-02	Maplecrest	Urban	A040	Candidate A040	Party Alpha		31446
2022	18/06/2022	Highmoor	Northern	HMR-03	HawkrIDGE	Semi-Urban	A008	Candidate A008	Party Alpha		26457
2022	18/06/2022	Northaven	Northern	NHV-03	Cedar Point	Rural	D003	Candidate D003	Party Delta		3942
2022	18/06/2022	Highmoor	Northern	HMR-03	HawkrIDGE	Semi-Urban	B008	Candidate B008	PARTY BETA		31115
2026	20/06/2026	Eastmere	Eastern	emr-03	Marigold Downs	Rural	D012	Candidate D012	Party Delta		7425
2026	20/06/2026	Midfield	Central	MDL-03	Ivy Crossing	Urban	A036	Candidate A036	Party Alpha		27295
2026	06/20/2026	Central Ridge	Central	CRG-04	Orchard Rise	Semi-Urban	G036	Candidate G036	Party Gamma		11819
2022	18/06/2022	Southvale	Southern	SVL-04	Yarrow Heath	Rural	G021	Candidate G021	Party Gamma		5169
2022	18/06/2022	Highmoor	Northern	HMR-05	Juniper Flats	Rural	G009	Candidate G009	Party Gamma		4317
2026	20 Jun 2026	Highmoor	Northern	HMR-01	Fernhollow	Rural	B006	Candidate B006	Party Beta		9919
2026	2026-06-20	Highmoor	Northern	HMR-02	Glenmarsh	Rural	B007	Candidate B007	Party Beta		4745
2026	20/06/2026	Lakeshire	Southern	LKS-02	Copperfield	Semi-Urban	G024	Candidate G024	Party Gamma		16043
2022	18/06/2022	Southvale	Southern	SVL-02	Violet Fields	Rural	B023	Candidate B023	Party Beta		14528
2026	20/06/2026	Northaven	Northern	NHV-05	Elmstead	Urban	B005	Candidate B005	Party Beta		37960
2022	18/06/2022	Coastgate	Eastern	CGT-02	Quarry End	Rural	A018	Candidate A018	Party Alpha		15779
2022	18/06/2022	Coastgate	Eastern	CGT-04	Yarrow Heath	Rural	A025	Candidate A025	Party Alpha		48044

2.1 Step zero: keep the original

Never clean the only copy. Right-click the RawElectionResults tab → **Move or Copy** → tick **Create a copy**. Rename the copy **Clean_Work**. You can always return to the original to check what changed.

2.2 Convert the range into an Excel Table

An **Excel Table** turns a plain range into a structured object. Tables expand automatically, keep formatting consistent, copy formulas down the whole column and allow readable **structured references** such as **ElectionRaw[VotesReceived]** instead of **K2:K281**.

1. Click any cell inside the data.
2. Press **Ctrl + T** (or Insert → Table).
3. Confirm **My table has headers** is ticked → OK.
4. On the **Table Design** tab, rename the table to **ElectionRaw** in the Table Name box.

	A010	Candidate A010	Party Alpha	
	D032	Candidate D032	Party Delta	
ban	I003	Candidate I003	Independent	
ban	A017	Candidate A017	Party Alpha	
ban	A01		Party Alpha	
	G01		Party Gamma	
	A01		Party Alpha	
	A02		Party Alpha	
ban	B04		Party Beta	
	G02		Party Gamma	
	B00		Party Beta	
	A01		Party Alpha	
	A02		Party Alpha	
	B019	Candidate B019	Party Beta	
ban	D019	Candidate D019	Party Delta	

?

×

Create Table

Where is the data for your table?

\$A\$1:\$M\$276 ↑

What is the name of your table?

Table1

☒ My table has headers

OK Cancel

2.3 Identify and remove duplicates

First **identify** duplicates so you know how many there are, then remove them. Add a helper column called **DupCount**:

HELPER COLUMN: DupCount

```
=COUNTIFS([ElectionYear],[@ElectionYear],[CandidateName],[@CandidateName],
[ConstituencyName],[@ConstituencyName],[VotesReceived],[@VotesReceived])
```

Any value greater than 1 means the record appears more than once. Filter DupCount > 1 to review them. Then remove them: **Data** → **Remove Duplicates** → tick every column except DupCount → OK. Excel reports how many duplicate rows it removed.

COMMON MISTAKE | Removing duplicates too early

If you run Remove Duplicates before fixing text problems, "Party Alpha" and "PARTY ALPHA" look different, so a genuine duplicate may survive. In our file the duplicates are exact copies, so the order does not matter here, but in real projects **standardise text first, then remove duplicates**.

2.4 Find & Replace

Use **Ctrl + H** for fast, consistent fixes across the whole table:

Find	Replace with	Options
NULL	(leave empty)	Tick Match entire cell contents
N/A	(leave empty)	Tick Match entire cell contents
-	(leave empty)	Select the PreviousElectionVotes column first; tick Match entire cell contents
Northavan	Northaven	Match case not needed

Find	Replace with	Options
Eastmeer	Eastmere	
Lakeshir	Lakeshire	Tick Match entire cell contents so 'Lakeshire' is not changed to 'Lakeshiree'

TRAINER TIP | Fix state names by lookup, not by typing

State and Region depend on ConstituencyCode. Once codes are fixed (Section 2.8), you can overwrite State with a lookup from Constituency Master, which corrects every spelling error at once. This is the approach we use in Module 4.

2.5 TRIM, CLEAN and SUBSTITUTE

Function	What it does	Example
TRIM(text)	Removes leading/trailing spaces and reduces internal repeated spaces to one	=TRIM(" Brightwater ") → "Brightwater"
CLEAN(text)	Removes non-printing characters (line breaks, tabs from copied systems)	=CLEAN(A2)
SUBSTITUTE(text, old, new)	Replaces every occurrence of old with new	=SUBSTITUTE("12,345",",","") → "12345"
PROPER / UPPER / LOWER	Changes letter case	=PROPER("party alpha") → "Party Alpha"

Combine them in one helper column to clean constituency names:

HELPER COLUMN: ConstituencyName_Clean
=TRIM(CLEAN([@ConstituencyName]))

2.6 Standardise party names with a mapping table

PROPER alone cannot fix "PartyAlpha" (no space) or "Indep.". The robust method is a **mapping table**: create a normalised key, then look up the official name.

Step 1. On a new sheet, create a small table named **PartyMap**:

Key	PartyName
PARTYALPHA	Party Alpha
PARTYBETA	Party Beta
PARTYGAMMA	Party Gamma
PARTYDELTA	Party Delta
INDEPENDENT	Independent
INDEP.	Independent

Step 2. In ElectionRaw, add a helper column **PartyName_Clean**:

HELPER COLUMN: PartyName_Clean

```
=XLOOKUP(UPPER(SUBSTITUTE([@PartyName], " ", "")), PartyMap[Key], PartyMap[PartyName], "CHECK")
```

Removing all spaces and converting to upper case turns every variant into the same key: " Party Alpha", "PARTY ALPHA", "PartyAlpha" and "party alpha" all become **PARTYALPHA**. Anything unexpected returns **CHECK**, so nothing is silently wrong.

2.7 LEFT, RIGHT and MID

These functions extract part of a text value. They are ideal for codes that follow a pattern.

Formula	Input	Result	Use
=LEFT(A2,3)	NHV-01	NHV	State code
=RIGHT(A2,2)	NHV-01	01	Constituency number
=MID(A2,8,3)	NHV-01-S03	S03	Station number
=RIGHT([@CandidateName],4)	Candidate A001	A001	Recover a missing CandidateID
=LEFT([@CandidateID],1)	B014	B	Check the party prefix

2.8 Standardise constituency codes

The raw file contains 16 codes written in different ways: **nhv-01**, **NHV01**, **NHV 01**, **NHV-1** and **NHV-01** (trailing space). All must become **NHV-01**.

HELPER COLUMN: ConstituencyCode_Clean

```
=LEFT(UPPER(TRIM([@ConstituencyCode])),3) & "-" & TEXT(VALUE(MID(SUBSTITUTE(SUBSTITUTE(UPPER(TRIM([@ConstituencyCode])), "-", "")), "", "")),4,2)), "00")
```

Read it from the inside out: TRIM removes spaces at the ends, UPPER fixes case, SUBSTITUTE removes hyphens and internal spaces (giving **NHV01** or **NHV1**), MID takes the digits after position 3, VALUE turns them into a number, and TEXT(...,"00") pads it back to two digits. The state code and a hyphen are added in front.

2.9 Flash Fill and Text to Columns

Flash Fill (Ctrl + E) learns a pattern from one or two examples. To recover missing CandidateIDs, type **A001** next to **Candidate A001**, type the next example, then press Ctrl + E. Always check a sample afterwards, because Flash Fill guesses.

Text to Columns (Data → Text to Columns) splits one column into several, or converts text to proper values. It is the quickest fix for mixed date formats:

1. Select the ElectionDate column.
2. Data → Text to Columns → Delimited → Next → untick all delimiters → Next.
3. Choose **Date: DMY** → Finish.
4. Check rows that were written month-first (06/18/2022) or in words; fix them by hand or rebuild the date from ElectionYear.

Because each election has one polling day, the most reliable fix is to rebuild the date from the year:

HELPER COLUMN: ElectionDate_Clean

```
=IF([@ElectionYear]=2022, DATE(2022,6,18), DATE(2026,6,20))
```

2.10 Numbers stored as text

15 VotesReceived values are text, such as "12,345" or " 9876". Text numbers are left-aligned and show a small green triangle. SUM ignores them without warning.

HELPER COLUMN: Votes_Clean

```
=IF([@VotesReceived]="", "", VALUE(SUBSTITUTE(TRIM([@VotesReceived]),",","")))
```

Test whether a cell is a real number with `=ISNUMBER([@Votes_Clean])`. Count remaining text values with `=SUMPRODUCT(--ISTEXT(ElectionRaw[VotesReceived]))`.

2.11 Missing values

Not every blank means the same thing. Decide case by case:

Missing field	Can it be recovered?	Action
ConstituencyName	Yes, from ConstituencyCode	Look up from Constituency Master (Module 4)
CandidateID	Yes, from CandidateName	=RIGHT([@CandidateName],4) or Flash Fill
VotesReceived	Yes, from polling-station data	Sum the party's station votes for that constituency; otherwise flag and exclude
PreviousElectionVotes (2026 rows)	Blank is correct when the party did not contest in 2022	Leave blank; do not replace with 0

COMMON MISTAKE | Replacing every blank with zero

A blank PreviousElectionVotes means *the party did not contest*, not *the party received zero votes*. Replacing it with 0 would make Vote Change % impossible to calculate and would falsely show huge increases.

2.12 Checking invalid values

Add a validation helper column that flags logical errors. One formula can test several rules:

HELPER COLUMN: VoteCheck

```
=IFS([@Votes_Clean]="", "Missing votes",
    [@Votes_Clean]<0, "Negative votes",
    [@Votes_Clean]>[@ValidVotes], "Votes exceed valid votes",
    TRUE, "OK")
```

Then reconcile each constituency. The sum of candidate votes should equal ValidVotes:

HELPER COLUMN: ReconDiff (should be 0)

```
=SUMIFS([Votes_Clean],[ElectionYear],[@ElectionYear],[ConstituencyCode_Clean],[@ConstituencyCode_Clean])
- [@ValidVotes]
```

A non-zero ReconDiff points to a missing, negative, duplicated or mistyped vote count in that constituency.

2.13 Data validation to prevent future errors

Cleaning fixes the past; **Data Validation** protects the future. Select a column → Data → Data Validation:

Column	Allow	Setting
PartyName	List	Source: =INDIRECT("PartyMap[PartyName]") or type the five names
VotesReceived	Whole number	Greater than or equal to 0
ElectionYear	List	2022, 2026
ConstituencyCode	Custom	=AND(LEN(A2)=6, MID(A2,4,1)="-")

Use **Data → Data Validation → Circle Invalid Data** to highlight existing values that break the rules.

HANDS-ON EXERCISE | Clean the raw election dataset (40 minutes)

Work on the Clean_Work sheet. Record what you find in a simple issue log (Issue, Rows affected, Fix applied).

1. Identify duplicates with a DupCount helper column, then remove them.
2. Identify missing values in ConstituencyName, CandidateID and VotesReceived.
3. Standardise party names using the PartyMap method.
4. Fix constituency codes to the SSS-NN format.
5. Convert VotesReceived into numeric format.
6. Check invalid values: negatives, votes above ValidVotes, and constituency totals that do not reconcile.
7. Prepare the dataset for analysis: replace original columns with cleaned values (Copy → Paste Values), delete helper columns you no longer need, and save as Clean_ElectionResults.xlsx.

CHECK YOUR ANSWER | What you should have found in RawElectionResults

- 6 duplicate records
- 5 blank constituency names and 4 missing CandidateIDs
- 34 inconsistent party names and 16 inconsistent constituency codes
- 15 vote counts stored as text, 3 missing vote counts and 2 negative values
- 1 vote count larger than the constituency's valid votes (an extra digit typed)
- 12 misspelt states, 10 names with extra spaces, 6 text placeholders (NULL, N/A, -) and 70 dates in non-standard formats

After cleaning you should have exactly **269 rows**: 136 for 2022 and 133 for 2026.

MODULE RECAP

- Keep the original; work on a copy converted to an Excel Table.
- Identify problems before fixing them, and log what you change.
- Use mapping tables and lookups for consistent standardisation.
- Check logic, not just format: votes must reconcile to valid votes.

MODULE 3

Excel Formulas for Election Analytics

From this module onward, use the clean file **Election Results.xlsx** (table name **ElectionResults**) and **Constituency Master.xlsx** (table name **ConstituencyMaster**). Your answers should match the *Check your answer* boxes exactly.

3.1 Basic aggregation functions

Function	Election example	Formula	Result
SUM	Total votes received, both elections	=SUM(ElectionResults[VotesReceived])	4,517,443
AVERAGE	Average 2026 turnout across constituencies	=AVERAGE(ConstituencyMaster[CurrentTurnoutPct])	76.04%
MAX	Highest 2026 turnout	=MAX(ConstituencyMaster[CurrentTurnoutPct])	85.38%
MIN	Lowest 2026 turnout	=MIN(ConstituencyMaster[CurrentTurnoutPct])	63.97%
COUNT	Number of numeric vote cells	=COUNT(ElectionResults[VotesReceived])	269
COUNTA	Number of candidate records (non-blank)	=COUNTA(ElectionResults[CandidateID])	269

COUNT counts only numbers; **COUNTA** counts any non-empty cell. On the raw file, COUNT(VotesReceived) is lower than COUNTA(VotesReceived) because some votes were stored as text. The gap is a quick data quality test.

3.2 Logical functions: IF, AND, OR, IFS and IFERROR

IF returns one value when a condition is true and another when it is false. **AND** requires all conditions to be true; **OR** requires at least one.

```
Result label:      =IF([@Rank]=1, "Won", "Lost")
Close winner flag: =IF(AND([@Rank]=1, [@Majority]<1000), "Close win", "")
Review flag:       =IF(OR([@CurrentTurnoutPct]<0.65, [@RejectedPct]>0.03), "Review", "")
```

IFS tests several conditions in order and returns the result of the first one that is true. It replaces long nested IF formulas.

High turnout classification

```
CONSTITUENCY MASTER: TurnoutBand
=IFS([@CurrentTurnoutPct]>=0.8, "High", [@CurrentTurnoutPct]>=0.65, "Medium", TRUE, "Low")
```

Victory classification

```
ELECTION RESULTS: VictoryBand
=IF([@Rank]<>1, "",
    IFS([@Majority]>10000, "Large Majority", [@Majority]>=5000, "Moderate Majority", TRUE,
    "Narrow Majority"))
```

IMPORTANT | These categories are arbitrary

The thresholds (80% / 65% for turnout; 10,000 / 5,000 for majority) are **training categories created for this exercise only**. They are not official definitions. In real work, agree thresholds with stakeholders and document them.

CHECK YOUR ANSWER | Classification results for 2026

- Turnout bands: High 11, Medium 28, Low 1 constituencies
- Victory bands: Large 14, Moderate 7, Narrow 19 constituencies

IFERROR replaces an error with a value you choose. It is essential when a denominator might be blank or zero:

```
=IFERROR([@VoteChange]/[@PreviousElectionVotes], "")
```

COMMON MISTAKE | Using IFERROR to hide real problems

Wrapping every formula in IFERROR can hide broken references and bad data. Use it only where you *expect* an error (such as no previous-election result) and know why it happens.

3.3 Conditional aggregation: SUMIF, SUMIFS, COUNTIF, COUNTIFS, AVERAGEIF, AVERAGEIFS

The *IF* family aggregates only the rows that meet criteria. The *IFS* versions accept multiple criteria and are preferred because the syntax is consistent: **result range first, then pairs of criteria range and criterion**.

Total votes by party (SUMIFS)

```
=SUMIFS(ElectionResults[VotesReceived],
        ElectionResults[PartyName], "Party Alpha",
        ElectionResults[ElectionYear], 2026)
```

Better practice: put the party and year in cells (for example H2 and H1) and reference them, so one formula can be copied down a summary table:

```
=SUMIFS(ElectionResults[VotesReceived], ElectionResults[PartyName], $G3,
        ElectionResults[ElectionYear], H$2)
```

Number of constituencies won (COUNTIFS)

```
=COUNTIFS(ElectionResults[PartyName], $G3, ElectionResults[ElectionYear], H$2,
            ElectionResults[Rank], 1)
```

Average turnout by state (AVERAGEIFS)

```
=AVERAGEIFS(ConstituencyMaster[CurrentTurnoutPct], ConstituencyMaster[State], "Highmoor")
```

CHECK YOUR ANSWER | Expected results

- Party Beta 2026 total votes: **914,455** (2022: 833,708); constituencies won 2026: **22** (2022: 19)
- Party Alpha 2026 total votes: **839,960** (2022: 833,677); constituencies won 2026: **16** (2022: 21)
- Party Gamma 2026 total votes: **419,325** (2022: 368,515); constituencies won 2026: **2** (2022: 0)
- Party Delta 2026 total votes: **149,977** (2022: 129,082); constituencies won 2026: **0** (2022: 0)
- Independent 2026 total votes: **13,613** (2022: 15,131); constituencies won 2026: **0** (2022: 0)

Average 2026 turnout, Highmoor: **80.82%**

Simple average vs weighted turnout

AVERAGEIFS gives every constituency equal weight. A large urban seat and a small rural seat count the same. The **weighted** turnout divides total ballots by total registered voters, so bigger constituencies count more. For Highmoor in 2026, the simple average is 80.82% but the weighted turnout is 80.10%. For Central Ridge: 71.55% vs 71.55%.

WEIGHTED TURNOUT

```
=SUMIFS(PollingStationResults[BallotsIssued], PollingStationResults[State], "Highmoor",  
PollingStationResults[ElectionYear], 2026)  
/ SUMIFS(PollingStationResults[RegisteredVoters], PollingStationResults[State], "Highmoor",  
PollingStationResults[ElectionYear], 2026)
```

BEST PRACTICE | Say which average you used

Both numbers are correct; they answer different questions. *Average constituency turnout* describes a typical seat. *Overall turnout* describes all voters together. Label charts and KPI cards clearly.

MODULE RECAP

- IF/IFS classify; AND/OR combine conditions; IFERROR handles expected errors.
- SUMIFS, COUNTIFS and AVERAGEIFS summarise by party, year, state and more.
- Reference criteria cells so formulas can be copied.
- Know the difference between a simple and a weighted average.

MODULE 4

Lookup Functions

Lookups bring information from one table into another using a shared key. In our dataset, **ConstituencyCode**, **CandidateID** and **PartyCode** are the keys that connect the tables.

4.1 Why master tables matter

A **master table** stores descriptive information once. Constituency Master stores each constituency's name, state, region and type in one row. The results table only needs the code; everything else can be looked up.

- **One source of truth:** fix a spelling once in the master, and every lookup updates.
- **Smaller fact tables:** results tables stay narrow and fast.
- **Consistency:** every report uses the same names and groupings.
- **Preparation for Power BI:** master tables become dimension tables in the data model (Module 15).

4.2 XLOOKUP

XLOOKUP searches one column and returns the matching value from another. It works left or right, returns exact matches by default and has a built-in *if not found* argument.

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])
```

Goal	Formula (in ElectionRaw or ElectionResults)
Candidate party from CandidateID	=XLOOKUP([@CandidateID], CandidateMaster[CandidateID], CandidateMaster[PartyName], "Not found")
Constituency name from code	=XLOOKUP([@ConstituencyCode], ConstituencyMaster[ConstituencyCode], ConstituencyMaster[ConstituencyName])
State (fixes misspellings)	=XLOOKUP([@ConstituencyCode], ConstituencyMaster[ConstituencyCode], ConstituencyMaster[State])
Registered voters	=XLOOKUP([@ConstituencyCode], ConstituencyMaster[ConstituencyCode], ConstituencyMaster[RegisteredVoters])
Constituency type	=XLOOKUP([@ConstituencyCode], ConstituencyMaster[ConstituencyCode], ConstituencyMaster[ConstituencyType])
PartyCode from clean party name	=XLOOKUP([@PartyName_Clean], PartyMaster[PartyName], PartyMaster[PartyCode])

4.3 Previous election result with multiple criteria

To find a party's 2022 votes in the same constituency, you need to match three things: year, constituency and party. Two approaches work:

OPTION A — SUMIFS (returns 0 if not found)

```
=SUMIFS([VotesReceived], [ElectionYear], 2022, [ConstituencyCode], [@ConstituencyCode], [PartyCode], [@PartyCode])
```

OPTION B — XLOOKUP with multiplied conditions (returns blank if not found)

```
=XLOOKUP(1, ([ElectionYear]=2022) * ([ConstituencyCode]=[@ConstituencyCode]) * ([PartyCode]=[@PartyCode]), [VotesReceived], "")
```

Option B is better here because it distinguishes *did not contest* (blank) from *received zero*. For Candidate A001 in Ashford Vale, both return **35,920**.

4.4 VLOOKUP

VLOOKUP is older but still common in existing workbooks, so you must be able to read it.

```
=VLOOKUP([@ConstituencyCode], ConstituencyMaster, 6, FALSE)
```

Limitation	Why it matters
Looks right only	The key must be the first column of the lookup range
Column number is hard-coded	Inserting a column in the master breaks the result silently (6 now points to the wrong field)
Approximate match by default	Forgetting FALSE can return a wrong value instead of an error

4.5 INDEX and MATCH

INDEX returns the value at a position; MATCH finds the position. Together they work in any direction and in every Excel version.

```
=INDEX(ConstituencyMaster[RegisteredVoters], MATCH([@ConstituencyCode],  
ConstituencyMaster[ConstituencyCode], 0))
```

A classic use is finding the name that goes with a maximum value:

```
HIGHEST-TURNOUT CONSTITUENCY  
=INDEX(ConstituencyMaster[ConstituencyName],  
MATCH(MAX(ConstituencyMaster[CurrentTurnoutPct]), ConstituencyMaster[CurrentTurnoutPct], 0))
```

Result: **Driftwood** (85.38%).

4.6 XMATCH

XMATCH returns a position like MATCH, with simpler defaults. It is useful for ranking checks, for example the position of a constituency in a list sorted by turnout:

```
=XMATCH("Ashford Vale", SortedTurnout[ConstituencyName])
```

COMMON MISTAKE | Looking up with dirty keys

A lookup fails if the key differs by a single space or letter case in some systems. **NHV-01** (trailing space) will not match **NHV-01**. Clean keys first (Module 2), and use the *if_not_found* argument so failures are visible.

HANDS-ON EXERCISE | Enrich the cleaned results

1. Fill the blank ConstituencyName values using XLOOKUP from Constituency Master.
2. Replace the State and Region columns with lookups, which corrects all misspellings.
3. Add a PartyCode column by looking up PartyName_Clean in Party Master.
4. Add RegisteredVoters and ConstituencyType.
5. Recalculate PreviousElectionVotes for 2026 rows using Option B and compare with the original column.

MODULE RECAP

- Master tables store descriptions once; keys connect tables.
- Prefer XLOOKUP; understand VLOOKUP; use INDEX/MATCH for compatibility.
- Multi-criteria lookups need all key parts: year, constituency and party.

MODULE 5

Election KPI Calculations

A **Key Performance Indicator (KPI)** is a number that summarises an important aspect of performance. In election analytics, a handful of KPIs answer most descriptive questions. This module defines each one, shows the Excel formula and works it through for **Ashford Vale (NHV-01), Election 2026**.

5.1 Turnout %

Definition: the share of registered voters who cast a ballot.

Turnout % = Ballots Issued / Registered Voters

Excel (Polling Station table): $=[@BallotsIssued]/[@RegisteredVoters]$
Constituency level: $=SUMIFS(PS[BallotsIssued], PS[ConstituencyCode], "NHV-01", PS[ElectionYear], 2026)$
 $/ SUMIFS(PS[RegisteredVoters], PS[ConstituencyCode], "NHV-01", PS[ElectionYear], 2026)$

Ashford Vale: $83,589 \div 118,954 = 70.27\%$

5.2 Vote Share %

Definition: the share of valid votes that a candidate or party received. Vote share uses **valid** votes as the denominator, not ballots issued and not registered voters.

Vote Share % = Candidate Votes / Total Valid Votes

Excel: $=[@VotesReceived]/[@ValidVotes]$

Candidate A001: $37,950 \div 82,291 = 46.12\%$

5.3 Rank

Rank orders candidates within the same constituency and election. Count how many candidates in the same contest received more votes, then add one:

$=COUNTIFS([ElectionYear], [@ElectionYear], [ConstituencyCode], [@ConstituencyCode], [VotesReceived], ">"&[@VoteReceived]) + 1$

5.4 Majority

Definition: the winner's votes minus the runner-up's votes. It is calculated for the winner only.

Majority = Winner Votes - Second Place Votes

Excel: $=IF([@Rank]=1, [@VotesReceived] - MAXIFS([VotesReceived], [ElectionYear], [@ElectionYear], [ConstituencyCode], [@ConstituencyCode], [Rank], 2), "")$

Ashford Vale: $37,950 - 22,417 = 15,533$

Majority is also often expressed as a percentage of valid votes, which makes large and small constituencies comparable: $\text{=[@Majority]/[@ValidVotes]}$ gives 18.88% for Ashford Vale.

5.5 Vote Change and Vote Change %

Vote Change = Current Election Votes - Previous Election Votes
Vote Change % = (Current Votes - Previous Votes) / Previous Votes

Excel: $\text{=IF([@PreviousElectionVotes]="", "", [@VotesReceived]-[@PreviousElectionVotes])}$
 $\text{=IFERROR([@VoteChange]/[@PreviousElectionVotes], "")}$

Candidate A001: $37,950 - 35,920 = 2,030$ votes, which is **5.65%**.

5.6 Vote Swing (percentage points)

Swing compares vote *share* between elections: 2026 share minus 2022 share. Party Alpha's share in Ashford Vale moved from 45.01% to 46.12%, a swing of **+1.11 pp**.

5.7 Rejected Vote %

Rejected Vote % = Rejected Votes / Ballots Issued

Excel: $\text{=[@RejectedVotes]/[@BallotsIssued]}$

Ashford Vale: $1,298 \div 83,589 = 1.55\%$. The overall 2026 rate is 1.68%.

5.8 Votes, share, turnout, majority and change: know the difference

Measure	Unit	Answers the question	Common confusion
Votes	Count	How many valid votes did the candidate get?	Larger constituencies naturally produce more votes
Vote Share	% of valid votes	What portion of the valid votes did the candidate get?	Using ballots or registered voters as the denominator
Turnout	% of registered voters	What portion of eligible voters cast a ballot?	Mixing up with vote share
Majority	Count (or %)	How far ahead was the winner?	Comparing raw majorities across very different seat sizes
Change	Count, % or pp	How did the value move since the last election?	Confusing % change with percentage-point change

COMMON MISTAKE | Percent vs percentage points

If turnout moves from 70% to 77%, it rose by **7 percentage points (pp)**, which is a **10% increase** ($7 \div 70$). Always use "pp" when subtracting two percentages.

HANDS-ON EXERCISE | Rebuild the KPI columns

1. In your cleaned Election Results table, add VoteShare, Rank, Majority, VoteChange and VoteChangePct columns using the formulas above.
2. In Polling Station Results, add Turnout and RejectedPct columns.
3. Filter to Ashford Vale 2026 and compare with the values in this module.
4. Filter VictoryBand = Narrow Majority. How many constituencies had a majority below 1,000 in 2026?

CHECK YOUR ANSWER | Narrowest 2026 majorities

- Quarry End (CGT-02): Party Beta by 156 votes
- Larchmont (EMR-02): Party Beta by 310 votes
- Granite Park (MDF-01): Party Gamma by 440 votes
- Amberly (SVL-05): Party Alpha by 529 votes
- Jasper Hollow (MDF-04): Party Alpha by 769 votes

MODULE 6

PivotTable for Election Analysis

A **PivotTable** summarises thousands of rows in seconds by dragging fields into Rows, Columns, Values and Filters. It is the fastest way to explore election data before building formulas or dashboards.

6.1 Creating a PivotTable

1. Click inside the ElectionResults table.
2. Insert → PivotTable → From Table/Range.
3. Choose **New Worksheet** → OK.
4. Drag fields from the PivotTable Fields pane into the four areas.

Area	Purpose	Election example
Rows	Groups displayed down the side	PartyName, State
Columns	Groups displayed across the top	ElectionYear
Values	Numbers being summarised	Sum of VotesReceived
Filters	Filter the whole PivotTable	ElectionYear = 2026

6.2 Votes by party

Rows: PartyName · **Values:** Sum of VotesReceived · **Filters:** ElectionYear = 2026. Sort descending: right-click a value → Sort → Largest to Smallest.

PartyName	Sum of VotesReceived (2026)	% of Grand Total
Party Beta	914,455	39.12%
Party Alpha	839,960	35.94%
Party Gamma	419,325	17.94%
Party Delta	149,977	6.42%
Independent	13,613	0.58%
Grand Total	2,337,330	100.00%

6.3 Show Values As: % of Grand Total

Drag VotesReceived into Values a second time. Right-click it → **Show Values As** → **% of Grand Total**. Rename the field to *Vote Share*. Because the PivotTable total equals all valid votes, this gives each party's national vote share.

6.4 Constituencies won by party

Rows: PartyName · **Values:** Count of ConstituencyCode · **Filters:** ElectionYear = 2026 and **Rank = 1**. Without the Rank filter you would count constituencies *contested*, not *won*.

PartyName	Won 2022	Won 2026	Contested 2026	Winning rate 2026
Independent	0	0	3	0.0%
Party Alpha	21	16	40	40.0%
Party Beta	19	22	40	55.0%

PartyName	Won 2022	Won 2026	Contested 2026	Winning rate 2026
Party Delta	0	0	22	0.0%
Party Gamma	0	2	28	7.1%

6.5 Turnout by state

Build this from **Constituency Master**. **Rows:** State · **Values:** Average of CurrentTurnoutPct (Value Field Settings → Average; Number Format → Percentage). Remember from Module 3 that this is the *simple* average of constituencies.

For a weighted turnout, build the PivotTable from **Polling Station Results** and add a **Calculated Field** (PivotTable Analyze → Fields, Items & Sets → Calculated Field): **Turnout = BallotsIssued / RegisteredVoters**. Calculated fields divide the *sums*, which gives the correct weighted figure.

State	Simple average 2026	Weighted turnout 2026
Highmoor	80.82%	80.10%
Southvale	78.53%	78.47%
Coastgate	77.04%	76.41%
Eastmere	76.35%	75.67%
Northaven	75.21%	74.70%
Lakeshire	75.65%	73.97%
Midfield	73.20%	72.23%
Central Ridge	71.55%	71.55%

6.6 Party performance by region

Rows: Region · **Columns:** PartyName · **Values:** Sum of VotesReceived · **Filter:** ElectionYear = 2026. Then Show Values As % of Row Total to see each party's share within each region.

Region	Alpha	Beta	Gamma	Delta	Independent
Central	35.2%	33.1%	27.4%	3.4%	0.9%
Eastern	33.6%	46.0%	12.6%	7.8%	—
Northern	45.0%	35.6%	12.6%	6.7%	—
Southern	30.0%	42.9%	17.4%	8.3%	1.4%

6.7 Election comparison

Rows: PartyName · **Columns:** ElectionYear · **Values:** Sum of VotesReceived. Add the field again and use Show Values As → **Difference From** → Base field ElectionYear, Base item 2022 to show the change.

PartyName	2022	2026	Difference	% Difference
Party Alpha	833,677	839,960	6,283	0.75%
Party Beta	833,708	914,455	80,747	9.69%
Party Gamma	368,515	419,325	50,810	13.79%
Party Delta	129,082	149,977	20,895	16.19%

PartyName	2022	2026	Difference	% Difference
Independent	15,131	13,613	-1,518	-10.03%

TRAINER TIP | Why total votes rose for most parties

Registered voters grew between 2022 and 2026 (2,991,487 → 3,171,746) and turnout moved from 74.02% to 74.95%. More votes overall does not automatically mean a larger *share*. Always check vote share alongside vote counts.

6.8 Sorting, filters, Top N, slicers and PivotCharts

Feature	How	Election use
Sort	Right-click a value → Sort → Largest to Smallest	Rank parties or states
Label filter	Row Labels drop-down → Label Filters	Show only states beginning with 'C'
Top N	Row Labels drop-down → Value Filters → Top 10	Top 10 constituencies by turnout
Slicer	PivotTable Analyze → Insert Slicer	Click to filter by ElectionYear, State, Region, ConstituencyType
Report Connections	Right-click slicer → Report Connections	One slicer controls several PivotTables
PivotChart	PivotTable Analyze → PivotChart	Chart that updates with the PivotTable

HANDS-ON EXERCISE | PivotTable practice (30 minutes)

1. Create the five PivotTables in this module on separate sheets named PT_Votes, PT_Won, PT_Turnout, PT_Region and PT_Compare.
2. Add an ElectionYear slicer and connect it to all five PivotTables.
3. Use Top 10 to list the ten constituencies with the highest 2026 turnout.
4. Create one PivotChart from PT_Won.

CHECK YOUR ANSWER | Top 5 turnout constituencies, 2026

- Driftwood (LKS-03, Lakeshire): 85.38%
- Saltmarsh (CGT-04, Coastgate): 84.01%
- Juniper Flats (HMR-05, Highmoor): 83.82%
- Quarry End (CGT-02, Coastgate): 82.39%
- Glenmarsh (HMR-02, Highmoor): 81.73%

MODULE 7

Excel Election Dashboard

A dashboard brings the most important numbers and charts onto one screen. In this module you combine your PivotTables, PivotCharts and slicers into a one-page **Fictional Election Dashboard** in Excel.

7.1 Plan the layout first

Sketch the layout before building. A proven structure is: title and filters at the top, KPI cards in one row, main charts in the middle, detail charts at the bottom.

Zone	Content
Top band	Title: 'Fictional Election Dashboard'; slicers for Election Year, State, Region, Constituency Type
KPI row	Six cards: Total Registered Voters, Total Votes Cast, Overall Turnout %, Total Constituencies, Largest Majority, Highest Turnout Constituency
Main charts	Votes by Party (bar), Constituencies Won by Party (bar)
Detail charts	Turnout by State (column), Vote Share by Party (100% bar or doughnut), Top 10 Constituencies by Turnout (bar)

7.2 KPI cards that respond to slicers

Formulas like SUMIFS do not react to slicers. The simplest way to make KPI cards interactive is to build **small single-value PivotTables** connected to the same slicers, then link a text box to each value.

1. Create a PivotTable from Polling Station Results with only **Sum of RegisteredVoters** in Values. Place it on a hidden sheet called Calc.
2. Repeat for Sum of BallotsIssued and for the calculated field Turnout.
3. Connect every Calc PivotTable to the dashboard slicers (Slicer → Report Connections).
4. On the dashboard, insert a rounded rectangle, click it, type `=Calc!B4` in the formula bar and press Enter. The shape now shows the PivotTable value.

For cards that return a name, use a formula on the Calc sheet that reads from a sorted PivotTable, for example the first row of a Top 1 turnout PivotTable.

CHECK YOUR ANSWER | KPI values with Election Year = 2026 and no other filter

- Total Registered Voters: **3,171,746**
- Total Votes Cast (ballots issued): **2,377,243**
- Overall Turnout: **74.95%**
- Total Constituencies: **40**
- Largest Majority: **21,098** (Brightwater, Party Alpha)
- Highest Turnout Constituency: **Driftwood** (85.38%)

7.3 Chart selection

Chart	Recommended type	Why
Votes by Party	Horizontal bar, sorted	Easy to compare lengths; long labels fit
Constituencies Won by Party	Horizontal bar	Counts compare cleanly

Chart	Recommended type	Why
Turnout by State	Column chart with axis starting at 0%	Eight categories fit across; avoids exaggerating differences
Vote Share by Party	100% stacked bar or doughnut	Parts of a whole; limit to five categories
Top 10 Constituencies by Turnout	Horizontal bar, Top 10 filter	Ranked list reads top to bottom

7.4 Formatting for clarity

- Hide gridlines (View → untick Gridlines) and use a light ivory background shape.
- Use the five neutral party colours consistently on every chart.
- Remove chart clutter: delete legends that repeat axis labels, remove unnecessary gridlines, add data labels where useful.
- Align objects with Shape Format → Align; group related items.
- Hide PivotChart field buttons (PivotChart Analyze → Field Buttons → Hide All).
- Lock the layout: Review → Protect Sheet (allow use of slicers and PivotTables).

HANDS-ON EXERCISE | Challenge 1 preview

Build the dashboard, then use it to answer the questions in Challenge 1 (Part 6). Change the slicers to 2022 and note which KPIs changed the most.

MODULE RECAP | Part 1

- Clean data first; everything else depends on it.
- Use SUMIFS/COUNTIFS/AVERAGEIFS and lookups for precise calculations.
- PivotTables explore quickly; slicers make them interactive.
- A dashboard is planned, not decorated: KPIs, comparisons, then detail.

PART 2

Generative AI for Election Data Analytics

Generative AI can write formulas, suggest checks, explain errors and summarise findings in seconds. In Part 2 you will learn to prompt it well and, just as importantly, to verify everything it produces against the data.

Module 8 Understanding Generative AI

Module 9 Prompt Engineering

Module 10 AI for Excel

Module 11 AI-Assisted Election Analysis

MODULE 8

Understanding Generative AI

8.1 Key terms

Term	Plain-English explanation	Example in this programme
Artificial Intelligence (AI)	Computer systems that perform tasks normally requiring human intelligence	Any tool that recognises patterns or generates answers
Machine Learning (ML)	AI that learns patterns from data instead of following hand-written rules	A model that predicts turnout from voter profile (not covered in depth here)
Generative AI	AI that creates new content: text, code, images, tables	Writing a SUMIFS formula or a DAX measure
Large Language Model (LLM)	A generative AI model trained on large amounts of text to predict and produce language	The engine behind ChatGPT, Claude, Copilot and Gemini
Prompt	The instruction you give the AI	"Write a DAX measure for turnout"
Hallucination	A confident answer that is wrong or invented	AI stating a turnout figure that is not in the data

8.2 Common Generative AI assistants

Tool	Provider	Strengths for analysts
ChatGPT	OpenAI	General reasoning, formula and code generation, file analysis in supported plans
Claude	Anthropic	Long documents, careful explanations, code and data analysis, structured outputs
Microsoft Copilot	Microsoft	Built into Excel, Power BI and Microsoft 365; works with your organisation's files under its data protection terms
Gemini	Google	Integrated with Google Workspace (Sheets, Docs) and Looker Studio workflows

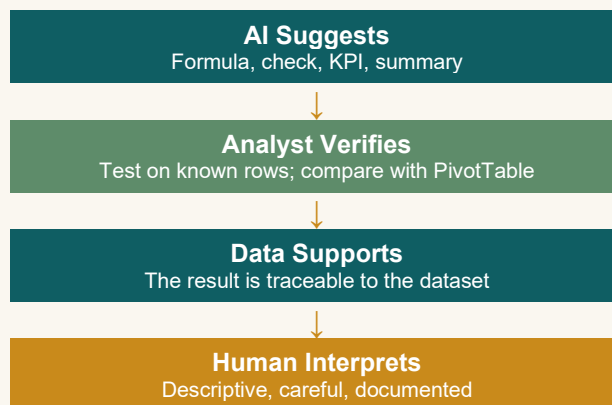
Features change quickly and depend on your licence. The prompting and verification skills in this part work with all of them.

8.3 What AI is good at, and what it is not

Good at	Weak at
Writing and explaining formulas, DAX and Power Query steps	Guaranteeing arithmetic accuracy on large pasted tables
Suggesting data quality checks and KPIs	Knowing your business rules unless you state them
Turning findings into clear summaries	Knowing whether your data is complete or correct
Brainstorming analytical questions	Explaining <i>why</i> people behaved in a certain way from numbers alone
Debugging error messages	Remembering context from earlier sessions unless provided

8.4 AI assists; it does not replace validation

AI output is a **draft**. Treat it like work from a fast but new colleague: useful, but checked before it goes into a report. The workflow in this programme is:



IMPORTANT | Data privacy and responsible use

- Our dataset is fictional, so it is safe for practice. With real organisational data, follow your organisation's AI policy.
- Do not paste personal, confidential or classified information into public AI tools.
- Prefer enterprise tools (for example Microsoft 365 Copilot under your organisation's licence) for work data.
- Often you only need to share **column names and a few sample rows**, not the full dataset.

MODULE RECAP

- LLMs generate language and code; they can be confidently wrong.
- Use AI to accelerate, then verify against data.
- Protect sensitive data; share structure, not secrets.

MODULE 9

Prompt Engineering

Prompt engineering is the skill of writing instructions that produce useful, accurate and well-formatted AI output. A good prompt removes guesswork. Use this six-part framework:

ROLE	TASK	CONTEXT	DATA	CONSTRAINT	OUTPUT FORMAT
------	------	---------	------	------------	---------------

Element	Question it answers	Election analytics example
ROLE	Who should the AI act as?	You are a data analyst experienced in Excel and Power BI.
TASK	What exactly should it do?	Identify the five constituencies with the highest and lowest turnout.
CONTEXT	What background does it need?	This is a fictional election dataset for a training course. Turnout = BallotsIssued ÷ RegisteredVoters.
DATA	What does it work on?	Columns: ElectionYear, ConstituencyCode, RegisteredVoters, BallotsIssued... [INSERT DATA]
CONSTRAINT	What limits apply?	Use only the supplied data. Do not speculate about voter motives. Show calculations.
OUTPUT FORMAT	How should the answer look?	A table with columns Constituency, State, Turnout %, sorted descending, then 3 bullet observations.

9.1 From poor to advanced

Poor prompt

AI PROMPT | Too vague

Analyse this election.

The AI does not know which election, which data, what 'analyse' means or what format you want. It may produce generic political commentary, which we specifically want to avoid.

Improved prompt

AI PROMPT | Specific task and data

Analyse this fictional election dataset and identify the three constituencies with the highest voter turnout.

Better: there is a clear task and a clear scope. But the AI still has to guess how turnout is defined and how to present the answer.

Advanced prompt

AI PROMPT | Full framework

You are a data analyst. Analyse the attached fictional election dataset. Compare voter turnout across states, identify the five constituencies with the highest and lowest turnout,

highlight unusual observations, and present your findings in a table. Do not make assumptions beyond the supplied data.

Verify: The five highest 2026 turnouts should be Driftwood, Saltmarsh, Juniper Flats, Quarry End, Glenmarsh. If the AI lists different names, check whether it calculated turnout correctly.

9.2 Techniques that improve results

Technique	How	Example
Give definitions	State formulas for KPIs	"Vote share = VotesReceived ÷ ValidVotes"
Show the structure	Paste headers and 5–10 sample rows	Copy the first rows of ElectionResults
Ask for steps	Request reasoning or a step list	"Explain each part of the formula"
Specify the format	Table, bullets, code block, word limit	"Return only the DAX code in a code block"
Iterate	Refine with follow-up prompts	"Now handle blanks in PreviousElectionVotes"
Ask it to check itself	Request assumptions and limitations	"List any assumptions you made"
Set boundaries	Say what not to do	"Do not speculate about why voters behaved this way"

COMMON MISTAKE | Asking AI to count or sum a huge pasted table

LLMs can make arithmetic slips when reading long pasted tables. Ask the AI for the **formula or method**, then let Excel or Power BI do the calculation. If you use an AI tool with a code-execution or data-analysis feature, still reconcile its totals with a PivotTable.

HANDS-ON EXERCISE | Upgrade a prompt (10 minutes)

Rewrite this weak prompt using all six elements: *"Which party did best?"* Hint: define 'best' (votes, vote share or constituencies won), specify the year, restrict the AI to the data and ask for a table.

MODULE RECAP

- ROLE → TASK → CONTEXT → DATA → CONSTRAINT → OUTPUT FORMAT.
- Define KPIs; share structure; ask for method over arithmetic.
- Iterate and set boundaries.

MODULE 10

AI for Excel

This module gives you 16 ready-to-use prompts for Excel tasks using the fictional election data. Replace the text in square brackets with your own details. After each prompt, a **Verify** note tells you how to check the output.

10.1 Formula generation

AI PROMPT | 1 · Generate a formula

I have an Excel table called ElectionResults. Write an Excel formula to calculate vote share using VotesReceived divided by ValidVotes. Use structured references and format the result as a percentage.

Verify: Expected: `=[@VotesReceived]/[@ValidVotes]`. Test on Candidate A001, Ashford Vale 2026: 46.12%.

AI PROMPT | 2 · Multi-criteria total

Write a SUMIFS formula that returns total VotesReceived for the party in cell G3 and the election year in cell H2 from the table ElectionResults. Explain each argument.

Verify: Party Alpha 2026 should return 839,960.

AI PROMPT | 3 · Rank within a constituency

In the ElectionResults table, write a formula that ranks each candidate within the same ElectionYear and ConstituencyCode by VotesReceived, where 1 is the highest. Do not use dynamic array functions.

Verify: Filter one constituency and confirm the winner shows 1 and ranks are consecutive.

AI PROMPT | 4 · Majority

Write an Excel formula for a Majority column in ElectionResults. For the Rank 1 candidate, return VotesReceived minus the votes of the Rank 2 candidate in the same ElectionYear and ConstituencyCode. For other rows return blank.

Verify: Ashford Vale 2026 should show 15,533 on the winner row.

AI PROMPT | 5 · Turnout band

Write an IFS formula that classifies CurrentTurnoutPct into High ($\geq 80\%$), Medium ($\geq 65\%$) and Low. These are training categories.

Verify: Count the bands: High 11, Medium 28, Low 1.

10.2 Troubleshooting

AI PROMPT | 6 · Formula returns zero

Explain why the following SUMIFS formula returns zero, and list the most likely causes in order of probability:

```
=SUMIFS(ElectionResults[VotesReceived], ElectionResults[PartyName], "Party Alpha", ElectionResults[ElectionYear], "2026")
```

Verify: Likely answers: trailing space in "Party Alpha ", year typed as text "2026" vs number, or votes stored as text. Test each fix.

AI PROMPT | 7 · Lookup returns #N/A

My XLOOKUP returns #N/A for some rows:

```
=XLOOKUP([@ConstituencyCode], ConstituencyMaster[ConstituencyCode], ConstituencyMaster[State]).
```

Sample failing keys are 'nhv-01', 'NHV01' and 'NHV-1'. Explain the cause and give a formula to standardise the keys to the format NHV-01.

Verify: Apply the formula to the 16 inconsistent codes and confirm every lookup succeeds.

AI PROMPT | 8 · Explain a formula

Explain this formula step by step for a beginner:

```
=LEFT(UPPER(TRIM(A2)),3)&"-"&TEXT(VALUE(MID(SUBSTITUTE(SUBSTITUTE(UPPER(TRIM(A2)),"-",""),",",""),4,2)), "00")
```

Verify: Compare with the explanation in Module 2, Section 2.8.

10.3 Data quality and validation

AI PROMPT | 9 · Data quality checks

Review the following fictional election dataset columns and recommend 10 data quality checks before analysis. For each check, give the Excel formula or tool to perform it. Columns:

[INSERT COLUMN LIST]

Verify: Your list should include duplicates, blanks, text numbers, negative values, votes exceeding valid votes and constituency reconciliation.

AI PROMPT | 10 · Validation rule

Suggest Excel validation rules to detect cases where VotesReceived exceeds ValidVotes. Provide both a Data Validation custom formula and a helper-column formula that flags existing violations.

Verify: The raw file contains one such row (an extra digit typed in Oakhaven, EMR-05, 2026). Your rule must catch it.

AI PROMPT | 11 · Reconciliation check

Write an Excel formula that checks, for each row in the table PollingStationResults, whether

ValidVotes + RejectedVotes equals BallotsIssued, and whether the five party vote columns sum

to ValidVotes. Return 'OK' or a short description of the problem.

Verify: Apply to RawPollingStations2026: three rows should fail the ballots check.

AI PROMPT | 12 · Clean party names

My PartyName column contains variants such as 'PARTY ALPHA', 'PartyAlpha', ' party alpha' and 'Indep.'. Suggest the most reliable Excel method to standardise them to 'Party Alpha', 'Party Beta', 'Party Gamma', 'Party Delta' and 'Independent'.

Verify: A good answer recommends a mapping table plus a normalised key; compare with Module 2.6.

10.4 KPIs, PivotTables and presentation

AI PROMPT | 13 · KPI suggestion

Based only on the following fictional election dataset, suggest appropriate descriptive KPIs

for an election analytics dashboard. For each KPI give the definition, the formula and the table it should come from. Columns: [INSERT COLUMN LIST]

Verify: Check that denominators are correct: turnout uses registered voters; vote share uses valid votes.

AI PROMPT | 14 · PivotTable design

I want to compare total votes by party between ElectionYear 2022 and 2026 in an Excel PivotTable, showing the difference and percentage difference. Tell me exactly which fields go in Rows, Columns and Values, and which Show Values As settings to use.

Verify: Party Beta should show 833,708 → 914,455.

AI PROMPT | 15 · Dashboard layout

Suggest a one-page Excel dashboard layout for a fictional election dataset with six KPI cards, five charts and four slicers (Election Year, State, Region, Constituency Type). Recommend a chart type for each visual and explain why.

Verify: Reject suggestions such as 3D charts or pie charts with many slices.

AI PROMPT | 16 · Weighted vs simple average

Explain the difference between the average of constituency turnout percentages and overall turnout calculated as total ballots divided by total registered voters. When should each be used on a dashboard?

Verify: Use Highmoor 2026 as your test: 80.82% (simple) vs 80.10% (weighted).

TRAINER TIP | Copilot in Excel

If your organisation has Microsoft 365 Copilot, open your Excel Table and use the Copilot pane to request formula columns, highlights or PivotTables in natural language. The same prompting framework applies, and so does verification.

MODULE 11

AI-Assisted Election Analysis

AI can help you move from *numbers* to *questions* and *findings*. This module shows how to use AI across the analytical workflow while keeping every statement traceable to the data.

11.1 Where AI helps in the workflow

Task	How AI helps	How you verify
Data quality checking	Suggests checks and formulas	Run the checks; count flagged rows
Descriptive analysis	Proposes summaries and groupings	Reproduce with a PivotTable
Ranking	Writes RANK, SORT or Top N logic	Sort the data manually and compare
Comparison	Designs year-on-year comparisons	Check two or three values by hand
Trend identification	Describes direction of change	Confirm with a chart of both years
Anomaly detection	Suggests rules (e.g. z-scores, thresholds)	Inspect each flagged record
Summarisation	Drafts plain-language findings	Every sentence must match a number
KPI recommendation	Suggests measures and definitions	Check denominators and grain
Formula / DAX creation	Writes code	Test on known results

11.2 Analytical questions the data can answer

Use AI to help calculate or structure answers to these questions. The expected answers from the training data are provided so you can verify the AI.

Question	Data-supported answer (2026 unless stated)
Which constituencies recorded the highest turnout?	Driftwood 85.38%; Saltmarsh 84.01%; Juniper Flats 83.82%
Which recorded the lowest turnout?	Eastlake 63.97%; Newbridge 67.32%; Ivy Crossing 67.39%
Which party received the highest total votes?	Party Beta: 914,455 (39.12% of valid votes)
Which constituencies had the narrowest margins?	Quarry End 156; Larchmont 310; Granite Park 440
Where did vote share change the most, 2022 to 2026?	Largest rise: Brightwater (Party Alpha, +23.0 pp); largest fall: Copperfield (Party Alpha, -25.1 pp)
Which regions show the greatest turnout change?	Central +0.9 pp; Eastern +1.1 pp; Northern +0.7 pp; Southern +1.0 pp
How many constituencies changed winning party?	6 of 40: Quarry End, Yarrow Heath, Driftwood, Granite Park, Lindenfield, Maplecrest

11.3 Anomaly detection with AI

An **anomaly** is a value that is unusual compared with the rest of the data. Anomalies are not necessarily errors, but they deserve investigation. Two examples are built into the training data:

Observation	Value	Comparison
Rejected vote rate, Redcliff (CGT-03), 2026	4.98%	National rate 1.68%; next highest 2.00%
Turnout change, Eastlake (LKS-04)	70.39% → 63.97%	-6.4 pp; the largest fall of all 40 constituencies

AI PROMPT | Anomaly rules

You are a data analyst. For a fictional election dataset with columns ConstituencyCode, TurnoutPct2022, TurnoutPct2026, RejectedPct2026, suggest three simple, explainable rules to flag unusual constituencies (for example, values more than 2 standard deviations from the mean). Give Excel formulas for each rule. Do not speculate about causes.

Verify: Apply the rules. Redcliff and Eastlake should both be flagged.

BEST PRACTICE | Describe, don't explain motives

Write: *"Redcliff's rejected-vote rate (4.98%) was about three times the national rate."*

Do not write: *"Voters in Redcliff were confused or protesting."* The data cannot show why. Recommend further data checks instead, such as reviewing station-level records.

Do not ask AI to speculate about why real people voted in particular ways. Keep questions to what the dataset can support.

11.4 When AI gets it wrong: a verification example

Suppose you paste Constituency Master into an AI tool and ask: *"What was overall turnout in Central Ridge in 2026?"* A typical AI response averages the five CurrentTurnoutPct values and answers 71.55%.

That is the **simple average**. The overall (weighted) turnout from polling-station data is 71.55%. Neither number is 'wrong', but the AI chose a method you did not specify. The fix is a better prompt: *"Calculate overall turnout as total BallotsIssued ÷ total RegisteredVoters."*

11.5 AI verification checklist

Check	Question to ask yourself
Source	Is every number traceable to a table and filter?
Definition	Did the AI use the KPI definition I intended?
Grain	Did it avoid summing repeated values (such as ValidVotes per candidate row)?
Completeness	Did it use all rows, or only what fit in the chat?
Reproduction	Can I reproduce the result with a formula or PivotTable?
Language	Are statements descriptive, not speculative?

HANDS-ON EXERCISE | Challenge 2 preview

Write five prompts: one to check data quality, one to suggest KPIs, one to explain an Excel formula, one to summarise descriptive trends and one to generate analytical questions. Test each, then verify at least two answers in Excel.

MODULE RECAP | Part 2

- Structured prompts produce structured, useful output.
- Let AI write methods; let Excel compute; let people verify.
- Anomalies are flags for investigation, not conclusions.
- Stay descriptive: the data shows what, not why.

PART 3

Microsoft Power BI

Power BI connects to many tables at once, cleans them repeatably, relates them in a model and turns them into an interactive report. In Part 3 you will rebuild the election analysis as a four-page dashboard that anyone can explore.

Module 12 Introduction to Power BI

Module 13 Connecting Election Data

Module 14 Power Query

Module 15 Election Data Model

Module 16 DAX for Election Analytics

Module 17 Advanced Election Metrics

Module 18 Building an Election Analytics Dashboard

MODULE 12

Introduction to Power BI

Power BI Desktop is Microsoft's free report-building application for Windows. Reports are published to the **Power BI Service** (app.powerbi.com) for sharing, subject to your organisation's licences. Every Power BI project follows six stages:

CONNECT	TRANSFORM	MODEL	ANALYSE	VISUALISE	SHARE
---------	-----------	-------	---------	-----------	-------

Stage	Power BI feature	Applied to the election dataset
Connect	Get Data	Import five Excel files: results, constituencies, candidates, parties, polling stations
Transform	Power Query Editor	Clean party names, fix codes, convert text numbers, remove duplicates
Model	Model view, relationships	Build a star schema with two fact tables and four dimensions
Analyse	DAX measures	Turnout %, Vote Share %, Constituencies Won, Vote Change
Visualise	Report view, visuals, slicers	Executive, Constituency, Party and Turnout pages
Share	Publish to Power BI Service	Share the report or app with colleagues; schedule refresh

12.1 The Power BI Desktop interface

Area	What you do there
Report view	Build pages with visuals, slicers and text
Table view	Inspect loaded data and calculated columns
Model view	Create and manage relationships
Power Query Editor	Clean and shape data before loading (Home → Transform data)

Area	What you do there
Visualizations pane	Choose visual types and assign fields
Data pane	Browse tables, columns and measures
Filters pane	Apply visual-, page- or report-level filters

TRAINER TIP | Excel skills transfer directly

- Excel cleaning → Power Query (but repeatable on every refresh)
- XLOOKUP and master tables → relationships in the model
- SUMIFS/COUNTIFS → DAX measures that respond to filters
- PivotTables and slicers → report visuals and slicers

MODULE 13

Connecting Election Data

13.1 Check table structure before importing

Before clicking Get Data, open each file and confirm: one header row, no merged cells, no totals rows, one table per sheet, consistent data types per column. Our training files are already formatted as named Excel Tables, which Power BI recognises directly.

File	Excel table name	Will become
Election Results.xlsx	ElectionResults	FactElectionResults
Polling Station Results.xlsx	PollingStationResults	FactPollingStation
Constituency Master.xlsx	ConstituencyMaster	DimConstituency
Candidate Master.xlsx	CandidateMaster	DimCandidate
Party Master.xlsx	PartyMaster	DimParty
(created in Power BI)	–	DimElection

13.2 Import the files

1. Open Power BI Desktop → **Home** → **Get Data** → **Excel workbook**.
2. Browse to **Election Results.xlsx** → Open.
3. In the **Navigator**, tick the table **ElectionResults** (the table icon, not the sheet icon).
4. Click **Transform Data** (not Load) so you can review it in Power Query.
5. In Power Query, use **Home** → **New Source** → **Excel workbook** to add the other four files the same way.

The screenshot shows the Power Query Editor interface. The ribbon at the top includes 'Home', 'Transform', 'Add column', 'View', and 'Help'. The 'Home' tab is active, showing various options for data manipulation. The 'Queries' pane on the left shows a list of queries, with 'Customers' selected. The main area displays the 'Customers' table with columns: CustomerID, CompanyName, ContactName, ContactTitle, Address, City, Region, PostalCode. The 'Query settings' pane on the right shows the 'Properties' tab with 'Name' set to 'Customers' and 'Entity type' set to 'Custom'. The status bar at the bottom indicates '1 warning', 'Columns: 13', and 'Rows: 91'.

BEST PRACTICE | Import vs DirectQuery

For Excel files, Power BI uses **Import** mode: data is copied into the report file and refreshed on demand. It is fast and supports all features. DirectQuery is used for large databases and is outside the scope of this programme.

BEST PRACTICE | Use a folder parameter

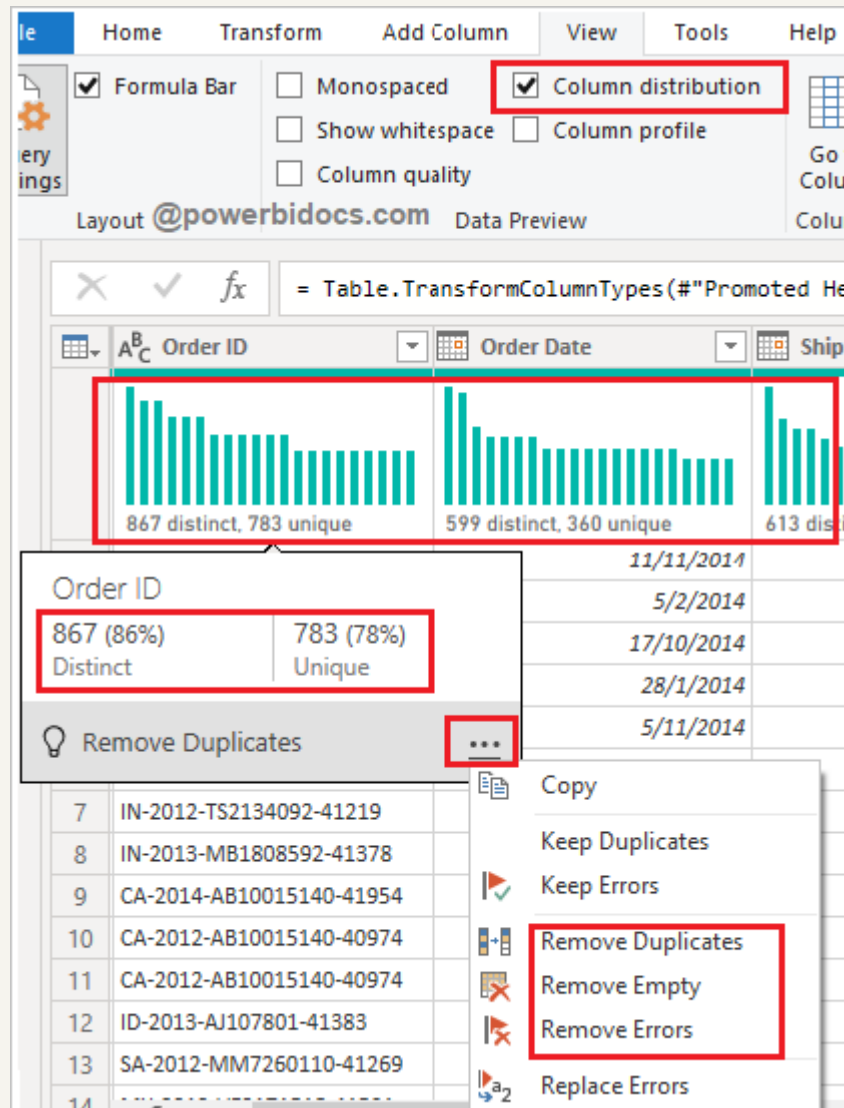
Store the file path in a parameter (Home → Manage Parameters) so the report still works when the folder moves. Change the parameter, refresh, done.

MODULE 14

Power Query

Power Query records every cleaning action as an **Applied Step**. When new data arrives, you click Refresh and every step runs again automatically. This is the biggest advantage over manual cleaning in Excel.

In this module you will clean **Raw Election Data (Messy).xlsx** in Power Query, repeating the Module 2 fixes in a repeatable way. Import both sheets: RawElectionResults and RawPollingStations2026.



14.1 Profile the data first

On the **View** tab, tick **Column quality**, **Column distribution** and **Column profile**. Change *Column profiling based on top 1000 rows* (status bar) to **entire data set**. Column quality shows the percentage of Valid, Error and Empty values for each column. Distribution shows distinct and unique counts, which reveals duplicates and inconsistent categories.

14.2 Core transformations

Transformation	Where	Election use
Rename columns	Double-click header	Rename ElectionYear columns consistently across tables
Change data types	Click the type icon on header	VotesReceived → Whole Number; ElectionDate → Date
Remove duplicates	Select key columns → Home → Remove Rows → Remove Duplicates	Remove duplicate candidate and station records
Replace values	Transform → Replace Values	NULL, N/A and - → null
Trim / Clean	Transform → Format → Trim / Clean	Remove extra spaces in names and codes
Remove blank rows	Home → Remove Rows → Remove Blank Rows	Drop empty rows in RawPollingStations2026
Split columns	Transform → Split Column → By Delimiter	Split PollingStationCode into constituency and station parts
Merge columns	Transform → Merge Columns	Create a key: ElectionYear & ConstituencyCode
Conditional column	Add Column → Conditional Column	TurnoutBand, VictoryBand
Merge Queries	Home → Merge Queries	Look up constituency names, states and party codes
Append Queries	Home → Append Queries	Stack 2022 and 2026 files into one table
Unpivot columns	Transform → Unpivot Columns	Turn five party vote columns into Party + Votes rows
Close & Apply	Home → Close & Apply	Load cleaned tables into the model

14.3 Standardise party names

The goal is to turn every variant into one official name:



Method 1 (user interface):

1. Select PartyName → Transform → Format → **Trim**.
2. Transform → Format → **UPPERCASE**.
3. Transform → Replace Values: replace a single space with nothing (this gives PARTYALPHA).
4. Replace **INDEP.** with **INDEPENDENT**.
5. Home → **Merge Queries**: match PartyName to a PartyMap query (Key column) and expand the official PartyName and PartyCode.

Method 2 (one custom column): Add Column → Custom Column:

POWER QUERY M — PartyName_Clean

```
let k = Text.Upper(Text.Remove([PartyName], {" "}))
in if k = "PARTYALPHA" then "Party Alpha"
   else if k = "PARTYBETA" then "Party Beta"
   else if k = "PARTYGAMMA" then "Party Gamma"
   else if k = "PARTYDELTA" then "Party Delta"
   else if k = "INDEPENDENT" or k = "INDEP." then "Independent"
   else "CHECK"
```

TRAINER TIP | Method 1 is easier to maintain

A mapping table can be edited by anyone and handles new variants without touching code. Use custom M only for quick, stable rules.

14.4 Fix constituency codes

POWER QUERY M — ConstituencyCode_Clean

```
let c = Text.Upper(Text.Trim([ConstituencyCode])),
    digits = Text.Select(c, {"0".."9"})
in Text.Start(c, 3) & "-" & Text.PadStart(digits, 2, "0")
```

Text.Select keeps only the digits, so **NHV01**, **nhv-01**, **NHV 01** and **NHV-1** all produce **01** or **1**, which Text.PadStart turns into **01**.

14.5 Convert text numbers and rejected votes

In RawElectionResults, select VotesReceived → Replace Values , with nothing → Trim → change type to Whole Number. In RawPollingStations2026, RejectedVotes contains entries like **45 votes** and **nil**:

POWER QUERY M — RejectedVotes_Clean

```
let t = Text.Lower(Text.Trim(Text.From([RejectedVotes])))
in if t = "nil" then 0 else Number.From(Text.Select(t, {"0".."9"}))
```

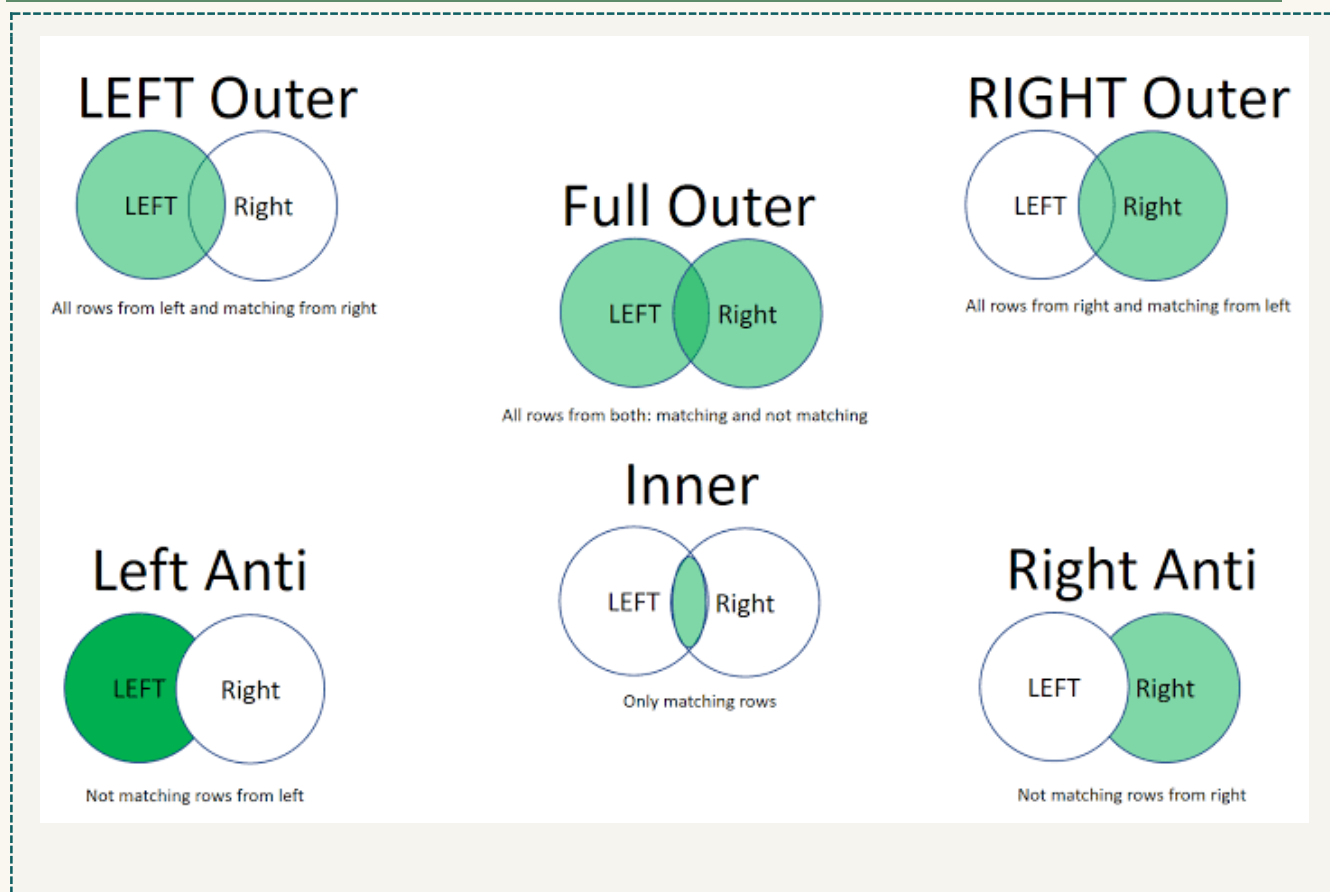
COMMON MISTAKE | Hiding type-conversion errors

When a type change produces **Error** cells, do not simply choose *Remove Errors*. That deletes whole records. First use *Keep Errors* to see which rows failed and why, fix the cause, then change the type.

14.6 Recover values with Merge Queries

Blank constituency names and misspelt states are fixed by merging with Constituency Master:

1. Select RawElectionResults → Home → **Merge Queries**.
2. Choose ConstituencyMaster as the second table.
3. Click ConstituencyCode_Clean in the first table and ConstituencyCode in the second. Join kind: **Left Outer**.
4. Confirm the match count at the bottom (all rows should match).
5. Expand the new column and tick ConstituencyName, State, Region and ConstituencyType. Remove the original messy columns.



14.7 Validation columns and filtering

Add a Conditional Column **VoteCheck**: if VotesReceived is null → "Missing"; if VotesReceived < 0 → "Negative"; if VotesReceived > ValidVotes → "Exceeds valid"; else "OK". Filter the preview to review non-OK rows. For training, correct them by referencing the clean file; in real projects, report them to the data owner.

14.8 Append and Unpivot

Append Queries stacks tables with the same columns. If polling-station results arrived as one file per election, you would append 2022 and 2026 into one table. Try it: append the cleaned RawPollingStations2026 to the 2022 rows of Polling Station Results (filter ElectionYear = 2022 first).

Unpivot converts wide party columns into a long format that works better with a party slicer:

Before (wide)	After (long)
PollingStationCode PartyAlphaVotes PartyBetaVotes ...	PollingStationCode Party Votes
NHV-01-S01 3,412 2,016 ...	NHV-01-S01 PartyAlphaVotes 3,412
	NHV-01-S01 PartyBetaVotes 2,016

Select the five party columns → Transform → **Unpivot Columns**. Then replace **PartyAlphaVotes** with **PA**, and so on, so the column can relate to DimParty. Name the new query **FactStationPartyVotes** (optional extension).

HANDS-ON EXERCISE | Clean the raw data in Power Query (35 minutes)

1. Profile both raw sheets using Column quality and distribution.
2. Standardise PartyName and ConstituencyCode.
3. Convert VotesReceived and RejectedVotes to whole numbers without losing rows.
4. Remove blank rows and duplicate records.
5. Merge with ConstituencyMaster to recover names and states.
6. Add VoteCheck and review every non-OK row.
7. Confirm the final row counts: 269 candidate records; 364 polling stations for 2026.

MODULE 15

Election Data Model

A **data model** is a set of tables connected by relationships. Power BI works best with a **star schema**: fact tables in the centre, surrounded by dimension tables.

Table type	Contains	Example
Fact table	Events or measurements: numbers you add up	FactElectionResults (VotesReceived), FactPollingStation (BallotsIssued)
Dimension table	Descriptions you filter and group by	DimConstituency (State, Region, Type), DimParty (PartyName)

15.1 The election star schema

Table	Key columns	Other columns
FactElectionResults	CandidateID, ConstituencyCode, PartyCode, ElectionYear	VotesReceived, ValidVotes, Rank, ResultStatus, Majority, PreviousElectionVotes, IncumbentStatus
FactPollingStation	PollingStationCode, ConstituencyCode, ElectionYear	RegisteredVoters, BallotsIssued, ValidVotes, RejectedVotes, party vote columns
DimConstituency	ConstituencyCode (unique)	ConstituencyName, State, Region, ConstituencyType, RegisteredVoters (2026 snapshot)
DimCandidate	CandidateID (unique)	CandidateName, AgeGroup, CandidateCategory
DimParty	PartyCode (unique)	PartyName, PartyGroup, ShortName
DimElection	ElectionYear (unique)	PreviousElectionYear, ElectionDate, ElectionLabel

15.2 Preparing the dimensions

DimCandidate must be unique

Candidate Master has one row per candidate **per election**, so a re-contesting candidate appears twice. A dimension key must be unique. In Power Query: duplicate the CandidateMaster query → keep CandidateID, CandidateName, AgeGroup, CandidateCategory → Remove Duplicates on CandidateID → rename to DimCandidate (161 rows).

IncumbentStatus changes by election, so it belongs in the fact table. Use **Merge Queries** on CandidateID **and** ElectionYear (hold Ctrl to select both columns) to bring IncumbentStatus into FactElectionResults.

Create DimElection

Home → **Enter Data** and type:

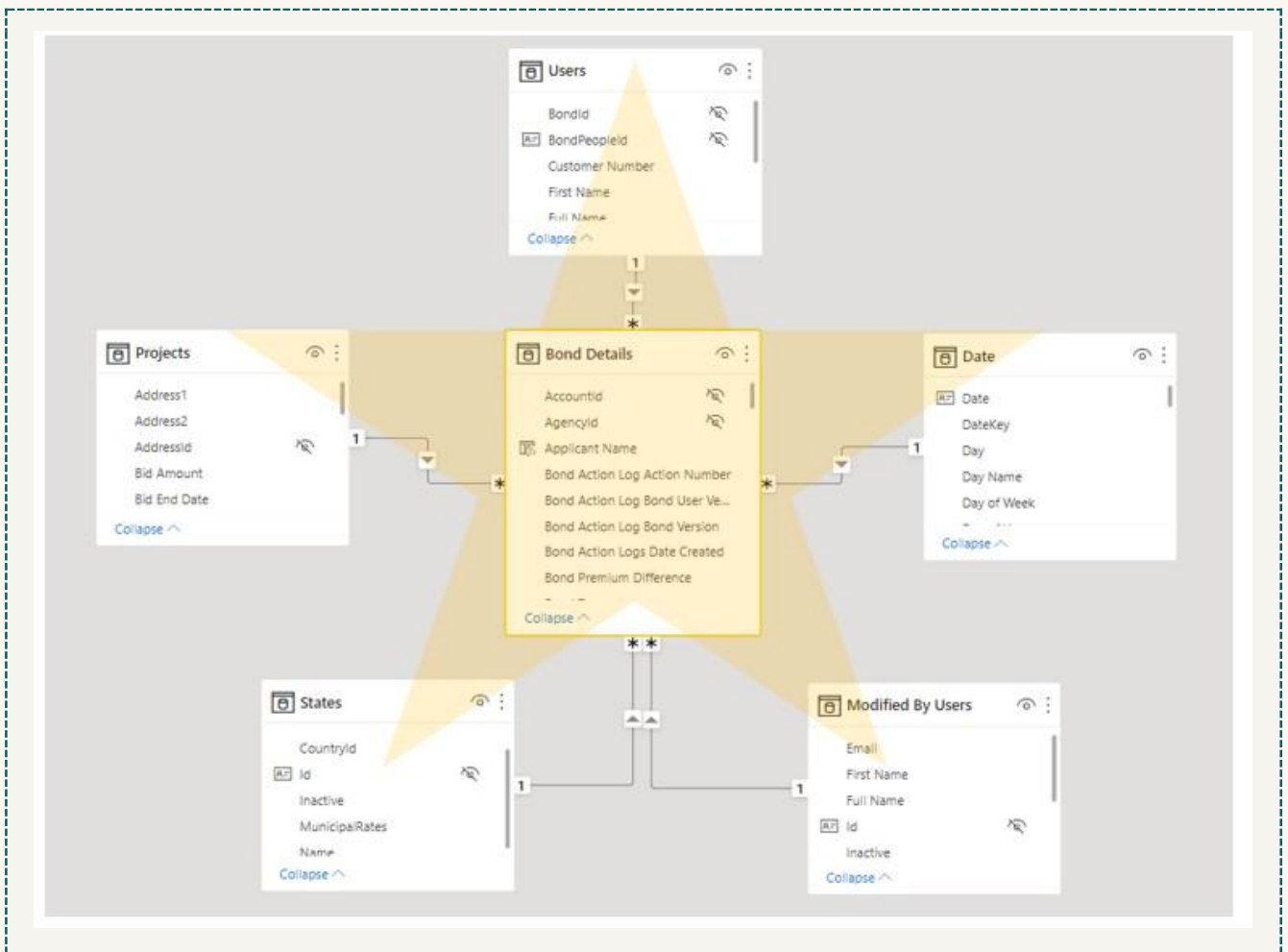
ElectionYear	PreviousElectionYear	ElectionDate	ElectionLabel
2022	(blank)	18/06/2022	Election 2022
2026	2022	20/06/2026	Election 2026

15.3 Relationships

In **Model view**, drag each key from the fact table onto the matching dimension key. Power BI usually detects them automatically; always check.

From (many side)	To (one side)	Cardinality	Filter direction
FactElectionResults[ConstituencyCode]	DimConstituency[ConstituencyCode]	Many-to-one	Single
FactElectionResults[CandidateID]	DimCandidate[CandidateID]	Many-to-one	Single
FactElectionResults[PartyCode]	DimParty[PartyCode]	Many-to-one	Single
FactElectionResults[ElectionYear]	DimElection[ElectionYear]	Many-to-one	Single
FactPollingStation[ConstituencyCode]	DimConstituency[ConstituencyCode]	Many-to-one	Single
FactPollingStation[ElectionYear]	DimElection[ElectionYear]	Many-to-one	Single

Single direction means filters flow from the dimension to the fact: selecting *Highmoor* in a State slicer filters both fact tables. Avoid bi-directional relationships unless you have a specific reason; they can create ambiguous filter paths.



15.4 Why two fact tables?

Turnout, valid votes and rejected votes belong to the **constituency or station**, not to a candidate. If they are stored only in the candidate table, they repeat for every candidate and summing them double counts,

as you saw in Module 1. Keeping them in FactPollingStation, with one row per station, means SUM is always correct.

The trade-off: FactPollingStation has no PartyCode, so a **Party slicer does not filter turnout**. This is correct behaviour; turnout is not a party-level measure. Explain this to report users with a note on the page.

COMMON MISTAKE | Using DimConstituency[RegisteredVoters] for turnout

DimConstituency holds a single 2026 snapshot of registered voters. It does not change when you select Election 2022. Always calculate registered voters from FactPollingStation, which has one value per station per election (2022: 2,991,487; 2026: 3,171,746).

MODULE RECAP

- Fact tables hold numbers; dimensions hold descriptions.
- Dimension keys must be unique; relationships are many-to-one.
- Store each number at its natural grain to avoid double counting.

MODULE 16

DAX for Election Analytics

DAX (Data Analysis Expressions) is the formula language of Power BI. A **measure** is a DAX calculation that is evaluated for whatever filters are active in a visual. The same measure shows national turnout in a card and state turnout in a chart.

16.1 Measure basics

- Create a dedicated table for measures: Home → Enter Data → name it `_Measures` → Load. Right-click it → New measure.
- Name measures in plain English with spaces: *Turnout %*, not *m_trn*.
- Use `DIVIDE(a, b)` rather than `a / b`; it returns blank instead of an error when b is zero.
- Format each measure (Measure tools → Format): whole number with thousand separator, or percentage with 2 decimals.

16.2 Core measures

Total Votes

```
Total Votes =
SUM ( FactElectionResults[VotesReceived] )
```

In plain English: Adds candidate votes for whatever is filtered. **Check:** 2026: 2,337,330

Registered Voters

```
Registered Voters =
SUM ( FactPollingStation[RegisteredVoters] )
```

In plain English: Sums registered voters from the station table, so each voter is counted once per election. Summing `DimConstituency[RegisteredVoters]` would ignore the election year; summing a candidate-level column would double count. **Check:** 2026: 3,171,746

Votes Cast

```
Votes Cast =
SUM ( FactPollingStation[BallotsIssued] )
```

In plain English: Ballots issued; the numerator of turnout. **Check:** 2026: 2,377,243

Total Valid Votes

```
Total Valid Votes =
SUM ( FactPollingStation[ValidVotes] )
```

In plain English: Valid votes from the station table. **Check:** 2026: 2,337,330

Total Rejected Votes

```
Total Rejected Votes =
SUM ( FactPollingStation[RejectedVotes] )
```

In plain English: Rejected ballots. **Check:** 2026: 39,913

Turnout %

```
Turnout % =
DIVIDE ( [Votes Cast], [Registered Voters] )
```

In plain English: Share of registered voters who cast a ballot. Because it divides two sums, it is automatically the weighted turnout at every level. **Check:** 2026: 74.95%

Vote Share %

```
Vote Share % =
DIVIDE (
    [Total Votes],
    CALCULATE ( [Total Votes], REMOVEFILTERS ( DimParty ), REMOVEFILTERS ( DimCandidate ) )
)
```

In plain English: Votes for the current party or candidate divided by all votes in the same area and year. REMOVEFILTERS clears only party and candidate filters, so State and Year filters still apply. **Check:** Party Alpha 2026: 35.94%

Total Constituencies

```
Total Constituencies =
COUNTROWS ( DimConstituency )
```

In plain English: Number of constituencies in the current State/Region/Type selection. **Check:** All: 40

Constituencies Won

```
Constituencies Won =
CALCULATE ( COUNTROWS ( FactElectionResults ), FactElectionResults[Rank] = 1 )
```

In plain English: Counts winning candidate rows. COUNTROWS (not DISTINCTCOUNT of codes) counts correctly across two elections. **Check:** Party Beta 2026: 22

Average Majority

```
Average Majority =
AVERAGE ( FactElectionResults[Majority] )
```

In plain English: Majority is blank on non-winning rows, and AVERAGE ignores blanks, so this averages winners only. **Check:** 2026: 7,466

Largest Majority

```
Largest Majority =
MAX ( FactElectionResults[Majority] )
```

In plain English: The biggest winning margin in the current selection. **Check:** 2026: 21,098

Narrowest Majority

```
Narrowest Majority =
MIN ( FactElectionResults[Majority] )
```

In plain English: The smallest winning margin. **Check:** 2026: 156

Previous Election Votes

```
Previous Election Votes =
VAR PrevYear = SELECTEDVALUE ( DimElection[PreviousElectionYear] )
RETURN
    IF (
        NOT ISBLANK ( PrevYear ),
        CALCULATE ( [Total Votes], REMOVEFILTERS ( DimElection ), DimElection[ElectionYear]
= PrevYear )
    )
```

In plain English: Finds the previous election year for the selected election, then recalculates Total Votes for that year, keeping all other filters (party, state). **Check:** Party Alpha, context 2026: 833,677

Vote Change

```
Vote Change =
IF ( NOT ISBLANK ( [Previous Election Votes] ), [Total Votes] - [Previous Election Votes] )
```

In plain English: Difference in votes compared with the previous election. **Check:** Party Alpha: 6,283

Vote Change %

```
Vote Change % =
DIVIDE ( [Vote Change], [Previous Election Votes] )
```

In plain English: Relative change in votes. **Check:** Party Alpha: 0.75%

COMMON MISTAKE | Implicit measures

Dragging VotesReceived straight into a visual creates an automatic (implicit) SUM. It works, but it cannot be reused, formatted centrally or referenced in other measures. Create explicit measures for everything you report.

MODULE 17

Advanced Election Metrics

This module builds a few more measures and explains **filter context**, the single most important DAX concept.

17.1 Filter context in plain English

Filter context is the set of filters that applies to a measure at the moment it is calculated. Filters come from slicers, from the rows and columns of a visual, from the Filters pane and from relationships.

Imagine a matrix with **PartyName** on rows and **ElectionYear** on columns, showing [Total Votes]. The cell for Party Beta / 2026 is calculated with two filters: DimParty[PartyName] = Party Beta and DimElection[ElectionYear] = 2026. The formula never changes; the filters do.

Visual cell	Active filters	Total Votes	Vote Share %
Party Beta, 2026	Party = Beta; Year = 2026	914,455	39.12%
Party Beta, 2022	Party = Beta; Year = 2022	833,708	38.24%
Total, 2026	Year = 2026	2,337,330	100.00%

CALCULATE is the function that changes filter context. In Vote Share %, the denominator uses CALCULATE with REMOVEFILTERS(DimParty) to remove the party filter, so it returns all votes for the year. That is why the share adds up to 100%.

TRAINER TIP | Test a measure in a table

When a measure gives an unexpected number, put it in a Table visual next to the fields that create its context (Year, State, Party). Seeing the value row by row usually reveals the problem.

17.2 Additional measures

Average Turnout (constituency average)

```
Average Turnout =
AVERAGEX ( VALUES ( DimConstituency[ConstituencyCode] ), [Turnout %] )
```

In plain English: Calculates Turnout % for each constituency, then takes the simple average. Use when you want the 'typical constituency', not all voters combined. **Check:** Highmoor 2026: 80.82% (vs Turnout % 80.10%)

Rejected Vote Rate

```
Rejected Vote Rate =
DIVIDE ( [Total Rejected Votes], [Votes Cast] )
```

In plain English: Share of ballots that were rejected. **Check:** 2026: 1.68%; Redcliff: 4.98%

Candidates Fielded

```
Candidates Fielded =
COUNTROWS ( FactElectionResults )
```


In plain English: Number of candidate records (one per constituency contested by a party). **Check:** Party Gamma 2026: 28

Winning Rate

```
Winning Rate =
DIVIDE ( [Constituencies Won], [Candidates Fielded] )
```

In plain English: Share of contests won by the party. **Check:** Party Beta 2026: 55.0%

Average Votes per Constituency

```
Avg Votes per Constituency =
DIVIDE ( [Total Votes], [Candidates Fielded] )
```

In plain English: Average votes in the constituencies the party contested. **Check:** Party Alpha 2026: 20,999

Vote Difference (2026 vs 2022)

```
Votes 2026 = CALCULATE ( [Total Votes], DimElection[ElectionYear] = 2026 )
Votes 2022 = CALCULATE ( [Total Votes], DimElection[ElectionYear] = 2022 )
Vote Difference = [Votes 2026] - [Votes 2022]
```

In plain English: Fixed-year comparison, useful for side-by-side columns regardless of the year slicer. **Check:** Party Gamma: 50,810

Turnout Change (pp)

```
Previous Turnout % =
VAR PrevYear = SELECTEDVALUE ( DimElection[PreviousElectionYear] )
RETURN CALCULATE ( [Turnout %], REMOVEFILTERS ( DimElection ), DimElection[ElectionYear] =
PrevYear )

Turnout Change pp = [Turnout %] - [Previous Turnout %]
```

In plain English: Change in turnout in percentage points. Format as a percentage and label it 'pp'. **Check:** National: +0.93 pp

Winning Majority %

```
Majority % =
DIVIDE ( SUM ( FactElectionResults[Majority] ), CALCULATE ( [Total Votes], REMOVEFILTERS (
DimParty ), REMOVEFILTERS ( DimCandidate ) ) )
```

In plain English: Majority as a share of valid votes, which makes seats of different sizes comparable. Use it at constituency level. **Check:** Ashford Vale 2026: 18.88%

COMMON MISTAKE | A party slicer and turnout

If a report user selects Party Alpha, Turnout % does not change, because FactPollingStation is not related to DimParty. This is correct. Add a subtitle such as *"Turnout is not filtered by party"* to avoid confusion.

MODULE RECAP | Modules 16–17

- Measures are evaluated in filter context.
- DIVIDE for safe ratios; CALCULATE to change filters; REMOVEFILTERS for shares.
- Every measure has a check value from the dataset; test before visualising.

MODULE 18

Building an Election Analytics Dashboard

You will now build the **Fictional Election Analytics Dashboard 2026**, a four-page report. Monthly or time-series visuals are not required because election results are event-based data: there are two election events, not a continuous timeline. Comparisons are made between election cycles instead.

18.1 Apply a report theme

Save the following as **ElectionTheme.json** and apply it with View → Themes → Browse for themes. It sets the ivory canvas, charcoal text and neutral party colours.

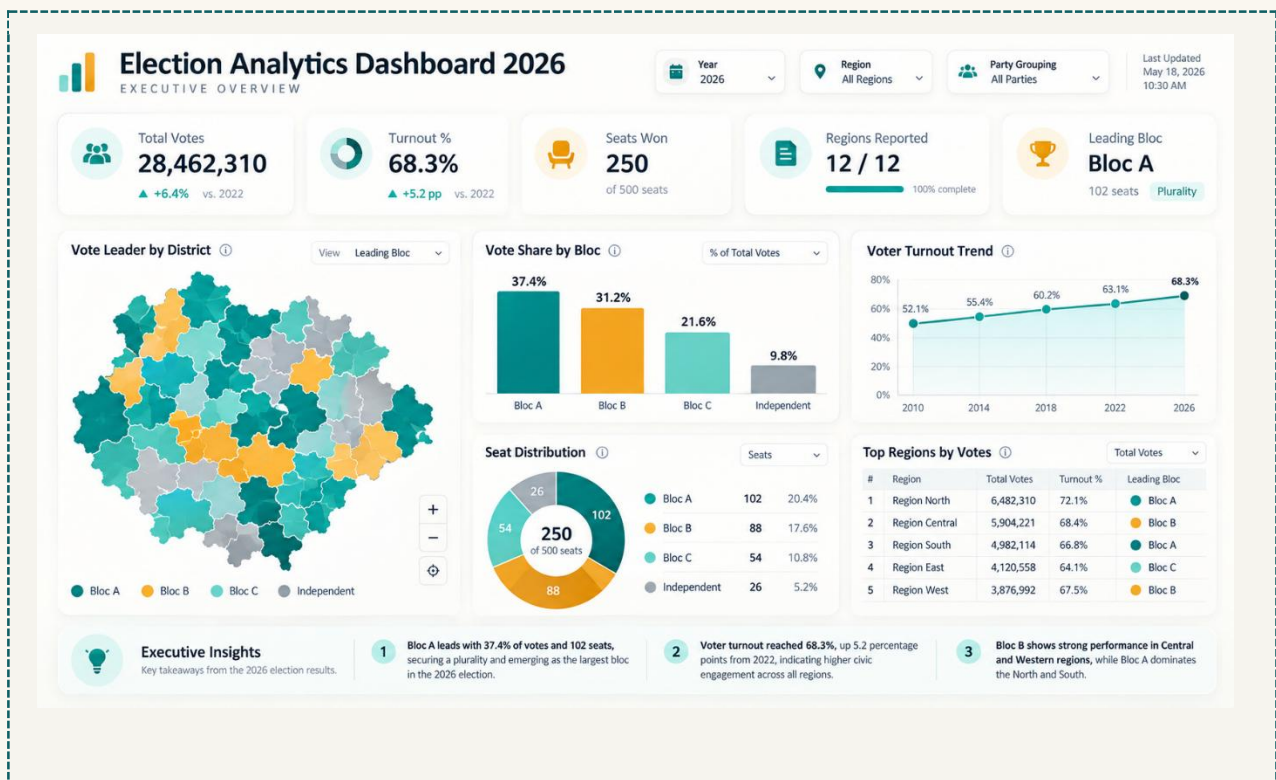
```
ElectionTheme.json
{
  "name": "Election Analytics Ivory",
  "dataColors": ["#3A7D7C", "#C98A1B", "#6C7A89", "#8C6A8F", "#9A9A9A", "#5E8C6A"],
  "background": "#FAF7F0", "foreground": "#2E2E2E", "tableAccent": "#0F5E63",
  "good": "#5E8C6A", "neutral": "#C98A1B", "bad": "#A5573A"
}
```

Also set fixed colours per party so they never change with sorting: select a party chart → Format → Bars/Columns → Colors → set each party to its colour from the case-study table.

18.2 Page 1 — Executive Overview

Element	Visual	Fields / measures
KPI cards (7)	Card (new) or multi-row card	Registered Voters, Votes Cast, Turnout %, Total Valid Votes, Total Rejected Votes, Total Constituencies, Average Majority
Constituencies Won by Party	Clustered bar	Y: DimParty[PartyName]; X: Constituencies Won; sort descending
Total Votes by Party	Clustered bar	Y: PartyName; X: Total Votes
Turnout by State	Clustered column	X: DimConstituency[State]; Y: Turnout %; Y-axis starts at 0
Turnout by Region	Clustered column	X: Region; Y: Turnout %
Vote Share by Party	100% stacked bar or donut	Values: Vote Share % by PartyName
Top Constituencies by Turnout	Bar, Top N filter = 10	Y: ConstituencyName; X: Turnout %
Narrowest Winning Margins	Table, Top N (bottom 10 by Narrowest Majority)	ConstituencyName, Winner party, Narrowest Majority
Election Year Comparison	Clustered column	X: PartyName; Y: Total Votes; Legend: ElectionYear
Slicers	Slicer (tile or dropdown)	Election Year, State, Region, Constituency Type, Party

Top N filter: select the visual → Filters pane → drag ConstituencyName into *Filters on this visual* → Filter type **Top N** → Show items Top 10 → By value: Turnout %. For narrowest margins choose **Bottom** 10 by Narrowest Majority, and add a visual-level filter Rank = 1.



18.3 Page 2 — Constituency Analysis

Element	Visual	Fields / measures
Constituency selector	Slicer, dropdown, single select	DimConstituency[ConstituencyName]
Profile cards	Cards	Registered Voters, Turnout %, Total Valid Votes, Rejected Vote Rate, Largest Majority (shows winner's majority)
Candidate ranking	Table	Rank, CandidateName, PartyName, Total Votes, Vote Share %; sort by Rank
Votes by candidate	Bar chart	Y: CandidateName; X: Total Votes; legend or colour by PartyName
Vote share	Donut	Vote Share % by PartyName
Previous election comparison	Clustered column	X: PartyName; Y: Total Votes; Legend: ElectionYear (remove Year slicer interaction)

Test with **Ashford Vale**: Turnout 70.27%, Candidate A001 46.12%, majority 15,533.

18.4 Page 3 — Party Performance

Element	Visual	Fields / measures
Party cards	Cards	Total Votes, Constituencies Won, Vote Share %, Avg Votes per Constituency, Winning Rate
Regional performance	Matrix with conditional formatting	Rows: Region; Columns: PartyName; Values: Vote Share %
State performance	Clustered bar	Y: State; X: Total Votes; filtered by Party slicer

Element	Visual	Fields / measures
Previous election comparison	Table	PartyName, Votes 2022, Votes 2026, Vote Difference, Vote Change %
Seats comparison	Clustered column	X: PartyName; Y: Constituencies Won; Legend: ElectionYear

18.5 Page 4 — Turnout Analysis

Element	Visual	Fields / measures
Overall turnout	Card / gauge	Turnout %, Turnout Change pp
Turnout by state	Column chart, sorted	State, Turnout %
Turnout by region	Column chart	Region, Turnout %
Urban vs Semi-Urban vs Rural	Column chart	ConstituencyType, Turnout %
Highest turnout constituencies	Bar, Top 10	ConstituencyName, Turnout %
Lowest turnout constituencies	Bar, Bottom 10	ConstituencyName, Turnout %
Turnout change between cycles	Table or scatter	ConstituencyName, Previous Turnout %, Turnout %, Turnout Change pp

Descriptive readings from the 2026 data: Rural constituencies recorded the highest turnout (79.85%), followed by Semi-Urban (77.08%) and Urban (71.54%). The largest fall in turnout was Eastlake (-6.4 pp); the largest rise was Umberton (+7.9 pp). Keep interpretation strictly descriptive: these are patterns in fictional data, not explanations.

18.6 Slicers, interactions and navigation

- **Sync slicers:** View → Sync slicers → tick the pages where Election Year and State should apply.
- **Edit interactions:** select a slicer → Format → Edit interactions → choose Filter or None for each visual. Set the Election Year slicer to *None* on 'Election Year Comparison' charts so both years stay visible.
- **Page navigator:** Insert → Buttons → Navigator → Page navigator.
- **Tooltips:** add Vote Share % and Majority to chart tooltips for extra detail without clutter.
- **Reset button:** Insert → Buttons → Reset, or a bookmark that clears slicers.

18.7 Design checklist

Check	Standard
Title	Every page has a clear title and a subtitle stating the data is fictional
Alignment	Visuals aligned to a grid; equal spacing
Colour	Party colours fixed and consistent; no colours of real political parties
Axes	Bar and column axes start at zero
Labels	Units shown (% , pp, votes); measures named in plain English
Clutter	No 3D effects; minimal gridlines; no unnecessary legends
Accuracy	KPI values match the check values in Modules 16–17

MODULE RECAP | Part 3

- Connect → Transform → Model → Analyse → Visualise → Share.
- Power Query makes cleaning repeatable.
- A star schema with correct grain prevents double counting.
- DAX measures plus consistent design produce a trustworthy dashboard.

PART 4

AI + Power BI

Generative AI is especially useful in Power BI, where DAX, Power Query M and model design have a steep learning curve. These prompts help you draft, debug and review, while you remain responsible for testing.

Part 4 18 reusable prompts for Power BI development

PART 4

Using AI to Accelerate Power BI Work

Each prompt below uses the fictional election model from Module 15. Replace bracketed placeholders where needed. After every AI answer: paste the code, check it runs, and compare the result with a check value from Modules 16–17.

AI PROMPT | 1 · DAX: vote share

You are a Power BI developer. I have a fictional election dataset containing CandidateID, PartyCode, ConstituencyCode, VotesReceived and ValidVotes in a table called FactElectionResults, with dimension tables DimParty and DimCandidate. Write a DAX measure for vote share that works at national, state and constituency level. Explain how filter context affects it.

Verify: Party Alpha 2026 = 35.94%.

AI PROMPT | 2 · KPI recommendation

You are a data analyst. Based only on the following fictional election dataset, suggest 8 descriptive KPIs for an executive dashboard. For each give a DAX measure and the table it should use. Tables: [INSERT TABLE NAME AND COLUMNS]

Verify: Reject KPIs that sum ValidVotes from the candidate-level table.

AI PROMPT | 3 · Model review

Review the following Power BI data model and identify possible relationship problems, ambiguous filter paths, or grain issues that could cause double counting. Model: [INSERT TABLES, KEYS AND RELATIONSHIPS]

Verify: The AI should flag a non-unique CandidateID in DimCandidate if you describe the raw Candidate Master.

AI PROMPT | 4 · Year comparison

Create DAX for comparing total votes between ElectionYear 2022 and ElectionYear 2026, using a DimElection table related to FactElectionResults. Provide measures for Votes 2022, Votes 2026, Difference and % Change.

Verify: Party Beta: 833,708 → 914,455.

AI PROMPT | 5 · Visual choice

Suggest an appropriate visual to compare turnout across 40 constituencies and explain your choice. Consider readability on a single report page.

Verify: A sorted bar chart with Top/Bottom N, or a small table with data bars, is a sensible answer; a 40-slice pie chart is not.

AI PROMPT | 6 · Previous election measure

Write a DAX measure that returns the previous election's Total Votes for the selected election, using `DimElection[PreviousElectionYear]`. It must keep party and constituency filters.

Verify: Party Alpha in 2026 context = 833,677.

AI PROMPT | 7 · Constituencies won

Write a DAX measure that counts constituencies won (`Rank = 1`) in `FactElectionResults`.

Explain why `COUNTROWS` may be safer than `DISTINCTCOUNT` when both election years are visible.

Verify: Party Beta: 2022 = 19, 2026 = 22, both years = 41.

AI PROMPT | 8 · Turnout without double counting

My turnout measure returns values above 100% when I use `FactElectionResults[ValidVotes]`. The

table has one row per candidate. Explain the cause and propose a correct model and measure.

Verify: The answer should mention grain and recommend a station- or constituency-level table.

AI PROMPT | 9 · Power Query M: clean codes

Write a Power Query M custom column that standardises constituency codes such as 'nhv-01', 'NHV01', 'NHV 01' and 'NHV-1' into the format 'NHV-01'.

Verify: Test on all 16 inconsistent codes in the raw file.

AI PROMPT | 10 · Power Query M: text numbers

Write Power Query steps to convert a `RejectedVotes` column containing values like '45 votes',

' 120 ', 'nil' and numbers into whole numbers, without deleting rows.

Verify: No row count should change after the step.

AI PROMPT | 11 · Unpivot

I have polling-station data with columns `PartyAlphaVotes`, `PartyBetaVotes`, `PartyGammaVotes`, `PartyDeltaVotes` and `IndependentVotes`. Explain how to unpivot these in Power Query and map them to `PartyCode` values PA, PB, PG, PD and IND.

Verify: Station totals must still equal `ValidVotes` after unpivoting.

AI PROMPT | 12 · Anomaly measure

Write a DAX measure that flags constituencies whose Rejected Vote Rate is more than twice the overall rate for the same election year. Return 'Review' or blank.

Verify: Redcliff (CGT-03) in 2026 must be flagged.

AI PROMPT | 13 · Top N dynamic title

Write a DAX measure for a dynamic chart title such as 'Top 10 Constituencies by Turnout - Election 2026 - Highmoor' that changes with the Election Year and State slicers.

Verify: Change slicers and confirm the title updates.

AI PROMPT | 14 · Explain a measure

Explain this DAX measure line by line for a beginner, including what CALCULATE and REMOVEFILTERS do: [INSERT DAX]

Verify: Compare with Module 17.1.

AI PROMPT | 15 · Debug DAX

This DAX measure returns the same value for every party in my bar chart: [INSERT DAX]. My model is [INSERT RELATIONSHIPS]. Diagnose the likely causes.

Verify: Typical causes: missing relationship, wrong column used on the axis, or REMOVEFILTERS on the wrong table.

AI PROMPT | 16 · Page layout

Design a layout for a Power BI 'Turnout Analysis' page for a fictional election dataset: list visuals, positions on a 16:9 canvas, and slicers. Keep interpretation descriptive only.

Verify: Check that each suggested visual can be built from existing measures.

AI PROMPT | 17 · Theme JSON

Create a Power BI theme JSON with an ivory background (#FAF7F0), charcoal text (#2E2E2E), teal accent (#0F5E63) and five neutral data colours suitable for fictional parties. Avoid colours associated with real political parties.

Verify: Apply it; confirm it loads without errors.

AI PROMPT | 18 · Report summary

You are a reporting analyst. Using only these measure results from a fictional election dashboard, write a 150-word descriptive executive summary without speculating about voter motives: [INSERT KPI VALUES]

Verify: Every number in the summary must appear in your dashboard.

TRAINER TIP | Copilot in Power BI

Depending on your organisation's capacity and licence, Copilot in Power BI can suggest measures, create report pages and summarise visuals. Treat its output exactly like any other AI draft: review measure definitions, check values and correct wording.

PART 5

End-to-End Capstone

You now bring everything together. Starting from raw, messy data, your team will produce a verified interactive dashboard and a short data-driven presentation.

Stage 1 Excel

Stage 2 AI

Stage 3 Power BI

Stage 4 Data Interpretation

CAPSTONE

From Raw Election Data to Interactive Dashboard

Work in groups of three or four. You receive **Raw Election Data (Messy).xlsx**, which contains deliberate quality issues in both sheets. Allow about 75 minutes, followed by 5-minute presentations per group.

Stage 1 — Excel

1. Convert the data into an Excel Table.
2. Clean party names.
3. Fix missing values.
4. Identify duplicates.
5. Calculate turnout (polling-station sheet).
6. Calculate vote share.
7. Create PivotTables.
8. Identify initial findings.

Stage 2 — AI

Use an AI assistant to:

1. Suggest data quality checks.
2. Review formulas.
3. Suggest KPIs.
4. Suggest useful analytical questions.
5. Explain unusual patterns (descriptively: what is unusual, not why).
6. Generate Excel formulas where required.

IMPORTANT | Verification is mandatory

Participants must verify all AI output. Keep a short log: prompt used, AI answer, how you verified it, and whether you accepted, corrected or rejected it.

Stage 3 — Power BI

1. Import the dataset.
2. Transform the data.
3. Create relationships.
4. Create DAX measures.
5. Build the dashboard.
6. Add slicers.
7. Configure interactions.

Stage 4 — Data Interpretation

Prepare a 5-minute presentation containing:

Deliverable	Quantity	Guidance
Important findings	5	Specific, numeric, traceable (e.g. 'Rural turnout was 8.3 pp higher than Urban')
Unusual observations	3	Outliers or anomalies with the comparison that makes them unusual
Data quality issues	3	What was wrong, how many rows, how you fixed it
Comparisons	3	Between years, regions, types or parties
Recommendations for further analysis	3	About data analysis, reporting or data quality only

IMPORTANT | Scope of recommendations

Recommendations must concern **data analysis, reporting or data quality**, for example 'collect station-level rejection reasons' or 'add a data validation step at entry'. They must not concern how voters should vote or how political campaigns should persuade voters.

Example findings format (from the training data)

- Overall 2026 turnout was 74.95%, +0.93 pp compared with 2022.
- 6 of 40 constituencies changed winning party between 2022 and 2026.
- Redcliff's 2026 rejected-vote rate (4.98%) was about 3.0 times the national rate (1.68%).
- 19 constituencies were won by fewer than 5,000 votes in 2026; the closest was Quarry End (156 votes).

Assessment rubric

Criterion	Weight	What good looks like
Data cleaning accuracy	25%	Correct row counts; all issue types addressed; issue log kept
Calculation correctness	25%	KPIs match check values; correct denominators; no double counting
Dashboard design	20%	Clear layout; required visuals and slicers; consistent colours
Use and verification of AI	15%	Structured prompts; documented verification
Interpretation	15%	Descriptive, specific, data-supported statements; appropriate recommendations

PART 6

Hands-On Challenges

Three timed challenges test your skills in each tool. Work individually first, then compare with a partner.

Challenge 1 Excel

Challenge 2 AI

Challenge 3 Power BI

PART 6

Challenge 1 — Excel

Using **Election Results.xlsx**, **Constituency Master.xlsx** and **Polling Station Results.xlsx**, find the following for **Election 2026**. Record the formula or PivotTable you used.

#	Question	Your answer	Method used
1	Total votes by party		
2	Average turnout by state		
3	Highest turnout constituency		
4	Lowest turnout constituency		
5	Largest winning majority		
6	Narrowest winning majority		

CHECK YOUR ANSWER | Challenge 1 answers

- Total votes: Party Beta 914,455; Party Alpha 839,960; Party Gamma 419,325; Party Delta 149,977; Independent 13,613
- Average turnout by state (simple / weighted): Highmoor 80.8% / 80.1%; Southvale 78.5% / 78.5%; Coastgate 77.0% / 76.4%; Eastmere 76.3% / 75.7%; Northaven 75.2% / 74.7%; Lakeshire 75.7% / 74.0%; Midfield 73.2% / 72.2%; Central Ridge 71.5% / 71.6%
- Highest turnout: Driftwood (LKS-03) 85.38%
- Lowest turnout: Eastlake (LKS-04) 63.97%
- Largest majority: Brightwater (Party Alpha) 21,098
- Narrowest majority: Quarry End (Party Beta) 156

Stretch questions

1. Repeat questions 1–6 for Election 2022. Which answers changed?
2. Which state had the largest gap between simple average and weighted turnout?
3. List the constituencies that changed winning party between elections.

PART 6

Challenge 2 — AI

Write your own prompts using the ROLE → TASK → CONTEXT → DATA → CONSTRAINT → OUTPUT framework. Test each one and record how you verified the result.

#	Prompt purpose	Your prompt (key parts)	Verified? How?
1	Check data quality		
2	Suggest KPIs		
3	Explain an Excel formula		
4	Summarise descriptive trends		
5	Generate analytical questions		

TRAINER TIP | Scoring your prompts

- 1 point for each framework element present (maximum 6).
- 1 point if the output was usable without major correction.
- 1 point for a documented verification step.

PART 6

Challenge 3 — Power BI

Create an **Election Executive Dashboard** containing:

Requirement	Minimum	Tick
KPI cards	5	☐
Charts	5	☐
Slicers	3	☐
Previous-election comparison	1	☐
Top N analysis	1	☐
All KPI values match check values	Yes	☐

PART 7

Prompt Library

Thirty-two reusable prompts organised by task. Replace the placeholders in square brackets with your own details.

Placeholders [INSERT DATA] [INSERT COLUMN] [INSERT TABLE NAME] [INSERT KPI] [INSERT ELECTION YEAR] [INSERT CONSTITUENCY]

PART 7

Reusable AI Prompt Library

All prompts assume a fictional dataset. For real organisational data, follow your data governance policy and share only what is necessary.

Data Cleaning

AI PROMPT | P01

You are a data analyst. Review the sample below from [INSERT TABLE NAME] and list every data quality problem you can see, grouped by type. Sample: [INSERT DATA]

AI PROMPT | P02

Suggest an Excel formula to standardise the values in [INSERT COLUMN] so that inconsistent spellings, spacing and capitalisation become one official value. Examples of variants: [INSERT DATA]

AI PROMPT | P03

Give me a step-by-step method to identify and remove duplicate records in [INSERT TABLE NAME], where a duplicate is defined by [INSERT COLUMN] values. Include how to count them before removal.

AI PROMPT | P04

The column [INSERT COLUMN] contains numbers stored as text, including commas and spaces. Give an Excel formula and a Power Query method to convert them to numbers without losing rows.

Excel Formulas

AI PROMPT | P05

Write an Excel formula using structured references for table [INSERT TABLE NAME] that calculates [INSERT KPI]. Define the KPI as [INSERT DEFINITION]. Explain each argument.

AI PROMPT | P06

Write a SUMIFS formula that totals [INSERT COLUMN] for ElectionYear [INSERT ELECTION YEAR] and the party named in cell [INSERT CELL].

AI PROMPT | P07

Explain why this formula returns [INSERT ERROR OR WRONG RESULT] and how to fix it: [INSERT FORMULA]

AI PROMPT | P08

Write an XLOOKUP that returns [INSERT COLUMN] from [INSERT TABLE NAME] using [INSERT KEY COLUMN], showing 'Not found' when there is no match.

PivotTable

AI PROMPT | P09

Tell me exactly how to set up a PivotTable from [INSERT TABLE NAME] to show [INSERT KPI] by [INSERT COLUMN] for [INSERT ELECTION YEAR], including Rows, Columns, Values, Filters and value settings.

AI PROMPT | P10

How do I show each party's percentage of the total in a PivotTable and compare [INSERT ELECTION YEAR] with the previous election using Show Values As?

AI PROMPT | P11

Explain how to calculate a weighted ratio such as [INSERT KPI] in a PivotTable using a calculated field, and why averaging percentages may be misleading.

Data Validation

AI PROMPT | P12

Suggest validation rules for [INSERT TABLE NAME] to ensure that RegisteredVoters $\geq$ BallotsIssued $\geq$ ValidVotes and ValidVotes + RejectedVotes = BallotsIssued. Provide helper-column formulas.

AI PROMPT | P13

Write a formula that checks whether the sum of candidate votes in [INSERT CONSTITUENCY] for [INSERT ELECTION YEAR] equals ValidVotes and returns the difference.

AI PROMPT | P14

Recommend Excel Data Validation settings for [INSERT COLUMN] to prevent invalid entries in future data collection.

Descriptive Analytics

AI PROMPT | P15

Based only on the data below, identify the five highest and five lowest values of [INSERT KPI] by constituency for [INSERT ELECTION YEAR]. Present a table. Do not speculate about causes. Data: [INSERT DATA]

AI PROMPT | P16

Compare [INSERT KPI] between the two election years for each region in the data below. Report the change in percentage points and rank regions by change. Data: [INSERT DATA]

AI PROMPT | P17

Suggest three explainable statistical rules to flag unusual values of [INSERT KPI], with Excel formulas. Do not interpret causes.

Power Query

AI PROMPT | P18

Write Power Query M to standardise [INSERT COLUMN] values such as [INSERT DATA] into the format [INSERT FORMAT].

AI PROMPT | P19

Explain how to merge [INSERT TABLE NAME] with a master table on [INSERT KEY COLUMN] in Power Query, which join kind to use, and how to check that all rows matched.

AI PROMPT | P20

Describe the Power Query steps to unpivot columns [INSERT COLUMN LIST] into attribute-value pairs and rename them to [INSERT NAMES].

DAX

AI PROMPT | P21

You are a Power BI developer. My model is [INSERT TABLES AND RELATIONSHIPS]. Write a DAX measure for [INSERT KPI] and explain how filter context affects it.

AI PROMPT | P22

Write DAX measures that compare [INSERT KPI] for [INSERT ELECTION YEAR] with the previous election using a DimElection table, including absolute and percentage change.

AI PROMPT | P23

This DAX measure returns [INSERT PROBLEM]: [INSERT DAX]. Diagnose the cause and correct it.

AI PROMPT | P24

Explain, line by line, what this DAX measure does for a beginner: [INSERT DAX]

Dashboard Design

AI PROMPT | P25

Recommend the best visual type for showing [INSERT KPI] across [INSERT NUMBER] categories, and explain why alternative visuals are less suitable.

AI PROMPT | P26

Design a one-page layout for an executive election dashboard with [INSERT NUMBER] KPI cards, charts and slicers. Give positions on a 16:9 canvas.

AI PROMPT | P27

Review this dashboard description for clarity, clutter and misleading design, and suggest improvements: [INSERT DESCRIPTION]

Data Interpretation

AI PROMPT | P28

Using only these results, write five descriptive findings, each beginning with the key number. Do not speculate about voter motives. Results: [INSERT DATA]

AI PROMPT | P29

From these KPI values, list three observations that are unusual compared with the rest of the data and state the comparison that makes each unusual: [INSERT DATA]

AI PROMPT | P30

Suggest three recommendations for further data analysis, reporting or data quality based on these findings: [INSERT FINDINGS]. Do not give political or campaign advice.

Executive Summary

AI PROMPT | P31

You are a reporting analyst. Write a 150-word executive summary of this fictional election dashboard for [INSERT ELECTION YEAR] using only these values: [INSERT KPI VALUES]. Use neutral, descriptive language.

AI PROMPT | P32

Convert these findings into five slide-ready bullet points of no more than 15 words each: [INSERT FINDINGS]

PART 8

Quick Reference

Cheat sheets to keep beside your keyboard after the programme.

Excel Key formulas

Metrics Election analytics definitions

AI Prompt framework

Power BI Workflow and common DAX

PART 8

Quick Reference Cheat Sheets

Excel

Function	Syntax	Election example
SUMIFS	=SUMIFS(sum_range, criteria_range1, criteria1, ...)	=SUMIFS(ER[VotesReceived],ER[PartyName],"Party Beta",ER[ElectionYear],2026)
COUNTIFS	=COUNTIFS(criteria_range1, criteria1, ...)	=COUNTIFS(ER[PartyName],"Party Beta",ER[Rank],1,ER[ElectionYear],2026)
AVERAGEIFS	=AVERAGEIFS(avg_range, criteria_range1, criteria1, ...)	=AVERAGEIFS(CM[CurrentTurnoutPct],CM[Region],"Northern")
IF	=IF(test, value_if_true, value_if_false)	=IF([@Rank]=1,"Won","Lost")
IFERROR	=IFERROR(value, value_if_error)	=IFERROR([@VoteChange]/[@PreviousElectionVotes],"")
XLOOKUP	=XLOOKUP(lookup, lookup_array, return_array, [not_found])	=XLOOKUP([@ConstituencyCode],CM[ConstituencyCode],CM[State],"Not found")

ER = ElectionResults table; CM = ConstituencyMaster table.

Election analytics metrics

Metric	Definition	Formula
Turnout	Share of registered voters who cast a ballot	Ballots Issued ÷ Registered Voters
Vote Share	Share of valid votes received by a candidate or party	Candidate Votes ÷ Valid Votes
Majority	Winning margin	Winner Votes – Runner-up Votes
Valid Votes	Ballots correctly marked and counted	Ballots Issued – Rejected Votes
Rejected Votes	Ballots not counted	Ballots Issued – Valid Votes; rate = Rejected ÷ Ballots Issued
Vote Change	Change in votes between elections	Current Votes – Previous Votes; % = change ÷ Previous

Metric	Definition	Formula
Swing	Change in vote share	Current Share – Previous Share (percentage points)

AI prompting

ROLE	TASK	CONTEXT	DATA	CONSTRAINT	OUTPUT
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Then: **AI Suggests** → **Analyst Verifies** → **Data Supports** → **Human Interprets**.

Power BI

CONNECT	TRANSFORM	MODEL	ANALYSE	VISUALISE	SHARE
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DAX: commonly used measures

```

Total Votes          = SUM ( FactElectionResults[VotesReceived] )
Registered Voters    = SUM ( FactPollingStation[RegisteredVoters] )
Votes Cast           = SUM ( FactPollingStation[BallotsIssued] )
Turnout %            = DIVIDE ( [Votes Cast], [Registered Voters] )
Rejected Vote Rate    = DIVIDE ( SUM ( FactPollingStation[RejectedVotes] ), [Votes Cast] )
Vote Share %         = DIVIDE ( [Total Votes],
                                CALCULATE ( [Total Votes], REMOVEFILTERS ( DimParty ),
                                REMOVEFILTERS ( DimCandidate ) ) )
Constituencies Won    = CALCULATE ( COUNTROWS ( FactElectionResults ),
FactElectionResults[Rank] = 1 )
Average Majority      = AVERAGE ( FactElectionResults[Majority] )
Votes 2026            = CALCULATE ( [Total Votes], DimElection[ElectionYear] = 2026 )
Winning Rate          = DIVIDE ( [Constituencies Won], COUNTROWS ( FactElectionResults ) )

```

Keyboard shortcuts

Shortcut	Action
Ctrl + T	Create Excel Table
Ctrl + H	Find & Replace
Ctrl + E	Flash Fill
Ctrl + Shift + L	Toggle filters
Alt + N + V	Insert PivotTable
F4	Toggle absolute/relative references
Alt + =	AutoSum

APPENDIX A

Dataset Generation Specification

This appendix specifies how the accompanying fictional Excel dataset was generated, so trainers can regenerate or extend it. All values are produced by a random process with a fixed seed; none are based on real elections.

A.1 Scale

Item	Specification	Delivered
States	8 fictional states	8 (Northaven, Highmoor, Eastmere, Coastgate, Southvale, Lakeshire, Midfield, Central Ridge)
Regions	4	Northern, Eastern, Southern, Central
Constituencies	40 (5 per state)	40
Parties	5 political categories	PA, PB, PG, PD, IND
Unique candidates	100–170	161
Election years	2	2022, 2026
Candidate-result records	200–400	269
Polling-station records	500–1,000	719

A.2 Generation logic

- Constituencies.** Each constituency is assigned a type. Registered voters are drawn uniformly from type ranges: Urban 85,000–120,000; Semi-Urban 55,000–85,000; Rural 32,000–58,000. 2026 registered voters = 2022 × a random growth factor of 3–9%.
- Contesting parties.** Party Alpha and Party Beta contest every seat. Party Gamma contests all Urban seats and about half of others. Party Delta contests about a third of seats, more often Rural. Independents appear in about 15% of contests.
- Vote shares.** Each region has base shares per party, adjusted by constituency type (for example Gamma +5 pp in Urban). Each constituency-party pair receives a persistent random factor (log-normal, $\sigma = 0.22$), and 2026 adds a smaller random change ($\sigma = 0.09$) plus small national shifts. Shares are normalised to sum to 1.
- Polling stations.** Each constituency has 6–12 stations (sometimes one more in 2026). Registered voters are split across stations with a Dirichlet distribution.
- Turnout.** Base turnout by type (Urban 71%, Semi-Urban 75%, Rural 79%) plus constituency noise; 2026 turnout = 2022 turnout + small random change. Station turnout varies around the constituency value and is capped between 45% and 93%.
- Ballots and votes.** BallotsIssued = RegisteredVoters × turnout. RejectedVotes = 0.8–2.4% of ballots. ValidVotes = BallotsIssued – RejectedVotes. Party votes at each station are drawn from a multinomial distribution on ValidVotes, so they sum exactly.
- Results.** Candidate VotesReceived = sum of the party's station votes. VoteSharePct = VotesReceived ÷ ValidVotes. Rank by votes; winner Rank = 1; Majority = winner – runner-up (winner row only).
- Cycle comparison.** For 2026 rows, PreviousElectionVotes = same party's 2022 votes in the same constituency; VoteChange and VoteSwingPct derived from them. About 93% of parties that contest both elections re-field the same CandidateID.

- 9. Deliberate teaching features.** Redcliff (CGT-03) has a 4–6% rejected-vote rate in 2026; Eastlake (LKS-04) has a 9-point turnout reduction applied; three constituencies are forced close between Alpha and Beta in 2026.

A.3 Integrity rules (all verified)

Rule	Status in clean files
Registered Voters $\geq$ Ballots Issued	Holds for every station
Ballots Issued $\geq$ Valid Votes	Holds for every station
Ballots Issued = Valid Votes + Rejected Votes	Holds exactly for every station
Candidate votes in a constituency sum to Valid Votes	Holds exactly
VoteSharePct = VotesReceived $\div$ ValidVotes	Holds (rounded to 4 decimals)
Winning candidate has Rank = 1	Holds
Majority = winner votes – runner-up votes	Holds
Station totals reconcile to candidate totals	Holds exactly

A.4 Messy file specification

Sheet	Injected issue	Count
Results	Blank constituency name	5
Results	Duplicate record	6
Results	Extra spaces	10
Results	Inconsistent constituency code	16
Results	Inconsistent date format	70
Results	Inconsistent party name	34
Results	Incorrect state spelling	12
Results	Missing CandidateID (recover from CandidateName)	4
Results	Missing vote count	3
Results	Negative value	2
Results	Number stored as text	15
Results	Text null placeholder (should be blank)	6
Results	Votes exceed ValidVotes (extra digit typo)	1
Stations 2026	BallotsIssued exceeds RegisteredVoters	2
Stations 2026	Blank row	3
Stations 2026	Duplicate polling station record	8
Stations 2026	Inconsistent state text	5
Stations 2026	Missing voter count	4
Stations 2026	Negative value	1
Stations 2026	Number stored as text	8
Stations 2026	Rejected votes recorded as text	10
Stations 2026	Valid + Rejected does not equal BallotsIssued	3

The raw results sheet omits calculated columns (PartyCode, VoteSharePct, Rank, Majority, VoteChange) so participants rebuild them. A row-by-row **IssueLog** with the correct value for each problem is provided in *Trainer Answer Key.xlsx*.

FINAL RECAP

10 Key Takeaways

- 1. Data quality comes first.** Every chart and KPI inherits the quality of the data beneath it. Garbage in, garbage out.
- 2. Excel helps you prepare and explore data.** Tables, cleaning functions, lookups and PivotTables turn raw records into reliable summaries.
- 3. Know your grain.** Store each number at its natural level (candidate, station, constituency) to avoid double counting.
- 4. Definitions matter.** Turnout uses registered voters; vote share uses valid votes; swing is measured in percentage points.
- 5. Weighted and simple averages answer different questions.** Label which one you show.
- 6. Generative AI helps you accelerate analytical tasks and problem-solving.** Use it for formulas, DAX, checks, questions and summaries.
- 7. AI suggests; you verify.** Test every AI output against known values before using it.
- 8. Power BI helps you model, visualise and communicate data.** Power Query makes cleaning repeatable; a star schema and DAX make analysis flexible.
- 9. Design for clarity.** Consistent colours, zero-based axes, clear titles and a logical page flow build trust.
- 10. Stay descriptive and evidence-based.** Report what the data shows; recommend further analysis rather than speculating.

EXCEL	+ AI	+ POWER BI	= END-TO-END DATA ANALYTICS
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BEST PRACTICE | A final reminder

Analytical conclusions must be supported by **reliable data**, **appropriate calculations** and **human validation**. Tools make analysis faster; people make it trustworthy.

Thank you for participating. For follow-up questions about the exercises, contact the trainer at mufasyahcons@gmail.com.

Fakhrul Syahmi · Data Consultant & HRD Corp Certified Trainer